Accurate Estimation of Small Effects: Illustration Through Air Pollution and Health

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Abstract

This paper identifies design parameters that can lead to inaccurate estimates of small effects. Low statistical power not only makes such effects difficult to detect but resulting significant estimates necessarily exaggerate true effect sizes on average. Through the literature on short-term health effects of air pollution, I explore this issue and its policy implications. Exaggeration can be substantial and power low even with large sample sizes. Real-data simulations highlight key additional drivers: the number of exogenous shocks, instrument strength, and outcome count. I propose a workflow to evaluate and mitigate exaggeration risk in non-experimental studies.

JEL codes: C18, C12, I18, Q53

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Project's website: https://vincentbagilet.github.io/inference_pollution

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Accurate estimates are central to credible economic analysis and public policy. Empirical findings routinely inform regulatory decisions, benefit-cost analyses, and resource allocation, so inaccurate estimates can lead to misguided policies and substantial welfare losses. Yet accurate estimation is challenging. In virtually all applied studies, if not the main effect, at least some effects of interest are relatively small, for instance when implementing heterogeneity analyses or exploring impacts on secondary outcomes. Such effects can be difficult to capture, even when the sample is large and identification is valid. When statistical power is low, significant estimates typically exaggerate the true effect, sometimes by large factors, rather than accurately reflect it (Gelman and Tuerlinckx 2000). Understanding which design features drive exaggeration in non-experimental settings is therefore essential for accurate estimation.

The present paper identifies tangible design parameters that can be acted on in individual studies to reduce this risk and explores the consequences of inaccurate estimates for public policy, through the lens of the literature on the short-term health effects of air pollution. This literature provides a particularly well-suited setting for this analysis. Its studies vary widely in identification strategies, effect sizes, populations of interest, and sample sizes considered. This heterogeneity makes it possible to evaluate, within a single framework, the many parameters and settings regularly encountered in applied economics, yielding insights that generalize well beyond this specific literature.

To identify concrete drivers of inaccurate estimation, I proceed in four steps. First, I illustrate through an example how low statistical power causes statistically significant estimates to exaggerate true effect sizes. Low power does not prevent obtaining a significant estimate, but such estimates typically exaggerate the true effect. When favored for publication, be it through editorial decisions or modeling and design choices, this carries over to published estimates (Gelman and Tuerlinckx 2000, Ioannidis 2008, Gelman and Carlin 2014). Power is the study-level driver of estimation biases resulting from selective publication. Second, I explore potential evidence of exaggeration in the literature on the short-term health effects of air pollution and assess implications for policy design. Third, I systematically identify drivers through Monte Carlo simulations using data from the US National Morbidity, Mortality, and Air Pollution Study (Samet et al. 2000), replicating common identification strategies: standard regression, reduced-form, instrumental variable, and regression discontinuity designs. Finally, I propose a workflow for evaluating exaggeration risks when conducting a non-experimental study.

In the literature analysis, I consider 2,692 estimates of the short-term health effects of air pollution from a unique corpus of 668 epidemiological studies and 36 articles relying on causal inference methods. The analysis reveals substantial heterogeneity in exaggeration risk. While some studies appear well-powered, a substantial share of estimates might be exaggerated, potentially by a large amount. These studies would not be adequately powered to detect and accurately measure plausible effects smaller than the point estimate they find. Yet an estimate is only informative if its associated design can detect a range of plausible effects, not only the point estimate the study obtained. For instance, an instrumental variable (IV) design can reasonably be expected to have enough power to detect effects of the magnitude of the corresponding naive OLS estimate, despite its potential bias. Half of the IV designs considered would have less than a 9% probability of detecting such a benchmark effect, and would exaggerate it by a factor of at least 4.5. A quarter would exaggerate it by a factor of at least 8.2. More generally, power issues appear despite otherwise reassuring design features: exaggeration arises even for large samples. I also find that less precise causal studies tend to report larger standardized effect sizes, a pattern consistent with selective publication and exaggeration. Finally, I briefly document a comparable heterogeneity in terms of exaggeration risk in the more general causal economics literature. Of course, any power analysis rests on a hypothesized true effect size and is informative only under that assumption. In the same spirit, my analyses do not aim to produce definitive exaggeration figures but should instead be regarded as ways to gauge the strength of each study's design.

Exaggeration has policy implications as academic estimates directly inform the computation of health benefits in regulatory impact analyses. Under arguably plausible exaggeration ratios, the marginal benefit-cost ratio of tightening the U.S. $PM_{2.5}$ standard from 10 to 9 $\mu g/m^3$ would fall from between 34 and 74 to between 6 and 14. While air pollution's large benefit-cost ratio preserves positive net benefits, exaggeration could lead to misallocation of resources across policies. In regulatory contexts with tighter benefit-cost margins, this could even reverse policy decisions. The accuracy and precision of these estimates are under direct scrutiny: in January 2026, the EPA announced it would no longer monetize the health benefits of reducing PM2.5 and ozone in Clean Air Act rulemakings, citing excessive uncertainty in existing estimates (U.S. EPA 2026). This decision further underlines the importance of understanding what drives imprecision in these estimates and should be an invitation to improve measurement rather than to abandon such critical quantification altogether.

The simulations identify concrete design features driving these patterns that likely extend to most settings beyond health or air pollution studies. I first show that, as expected, exaggeration increases when the sample size decreases. Substantial exaggeration can arise even for large sample sizes, on the order of 100,000 observations. Second, the simulations confirm that the smaller the effect targeted, the larger the exaggeration. Exaggeration can be substantial for effect sizes typically observed in the literature. Third, I find that using rare exogenous shocks can produce greatly inflated estimates even when sample and true effect sizes are large. Similarly, substantial exaggeration can arise when the instrument only explains a limited portion of the variation in air pollution, even when F-statistics are well beyond standard thresholds. Finally, the count of cases of the outcome is a key driver of exaggeration. As a consequence, estimated effects of air pollution on the elderly or children can be exaggerated due to the small number of daily hospital admissions or deaths for these groups. Overall, the simulations confirm that many determinants of exaggeration can be hidden behind features that appear protective, such as large sample sizes or large F-statistics.

This paper makes three main contributions. First, it contributes to the literature on the bias of published estimates arising from selective publication. The estimation biases resulting from publication bias, whose detection and correction are formalized for instance in Andrews and Kasy (2019), are conditional on low power and, at the study level, driven by it. Adequately powered studies would not lead to substantial exaggeration, even in the presence of selection on significance. The literature on publication bias typically focuses on one of the two ingredients for associated estimation biases, selective publication, and operates at the level of a body of literature to detect and correct for selection. What drives estimation biases at the level of individual studies, the second ingredient, remains largely uncharacterized. The present paper identifies the design features that, through their effect on statistical power, make a study prone to exaggeration. Existing meta-analyses document serious power issues in economics but typically do not identify their determinants (Ioannidis, Stanley and Doucouliagos 2017, Ferraro and Shukla 2020, Young 2022, Arel-Bundock et al. 2024). To my knowledge, no paper in economics systematically examines the concrete drivers of exaggeration in non-experimental designs. The closest existing work consists of simulation studies that document power issues in difference-in-differences settings (Schell, Griffin and Morral 2018, Griffin et al. 2021, Black et al. 2022). In particular, Black et al. (2022) provides a detailed methodological guide for conducting simulated power analyses in observational settings, illustrated through the case of health insurance and mortality. I complement this work by studying the concrete drivers of exaggeration in a wide array of research designs: standard regression, reduced-form, instrumental variable, and regression discontinuity designs. The present paper provides tangible, actionable guidance on these design parameters and discusses policy consequences of exaggeration. In a companion paper Bagilet (2026), I take a theoretical angle distinct from the empirical focus of the present paper and isolate the specific role of causal identification strategies in amplifying exaggeration by limiting the variation used for identification.

Second, I contribute to the literature on the replicability and credibility of empirical findings (Button et al. 2013, Open Science Collaboration 2015, Christensen and Miguel 2018, Camerer et al. 2016, Brodeur, Cook and Heyes 2020, Brodeur, Mikola and Cook 2024). Statistical power is often of interest in applied analyses before an effect is detected. Yet, exaggeration underlines that it is also critical after a statistically significant estimate has been obtained. I thus advocate reporting power calculations alongside published economics results to allow gauging exaggeration risks. To this end, the present paper is built so that it can be used as a template for such power calculations that can easily be adapted to a wide range of settings to assess exaggeration risk in individual studies. The section analyzing the literature provides an example of how one can conduct a retrospective power analysis, while the simulations illustrate how real or simulated data can be used to identify potential sources of exaggeration before undertaking a study. [1]

Finally, I contribute to the literature on the acute health effects of air pollution by systematically evaluating the reliability of published estimates across research designs and analyzing implications for regulatory policy. This literature is remarkably vast, with hundreds of studies published on this topic. It also directly informs environmental policy addressing a major public health concern. However, typically small underlying effect sizes and coarse data, with observations generally at the city-day level, make it particularly prone to exaggeration and power issues. I highlight risks of publication bias, low power, and exaggeration in this literature, show that certain designs can be more subject to these issues, and describe how exaggerated health estimates affect the air pollution regulatory process.

1 Background on Statistical Power and Exaggeration

Gelman and Tuerlinckx (2000) and Gelman and Carlin (2014) point out that statistically significant estimates suffer from a winner's curse in under-powered studies. These estimates can largely exaggerate true effect sizes or can even be of the opposite sign. In this section, I illustrate these two points through a simple simulation exercise. For clarity of exposition, this exercise builds on a short-term health effect of air pollution example.

1.1 Illustrative Setting

Let's simulate an "ideal" experiment in which a mad scientist is able to randomly increase the concentration of fine particulate matter ($PM_{2.5}$) to estimate the short-term effects of air pollution on daily non-accidental mortality. The experiment takes place in a major city over the 366 days of a leap year. The scientist increases the concentration of particulate matter by 10 $\mu g/m^3$, a large shock equivalent to a one standard deviation increase in the concentration of $PM_{2.5}$. Concretely, the scientist implements a complete experiment where they randomly allocate half of the days to the treatment group and the other half to the

[1] I make all replication and supplementary materials accessible and easily adaptable *via* the project's website.

control group. They then measure the treatment effect of the intervention by computing the average difference in means between treated and control outcomes. They find a treatment effect of 4 additional deaths that is statistically significant at the 5% level. The statistical significance of the estimate fulfills the scientist's expectations.

Contrary to the scientist, we know the true effect of the experiment since we created the data. Table 1 displays the pair of potential outcomes of each day, $Y_i(T_i = 0)$ and $Y_i(T_i = 1)$. $Y_i(T_i)$ represents the daily count of non-accidental deaths and T_i the treatment assignment, equal to 1 when unit i is treated and 0 otherwise. I first simulated the daily non-accidental mortality counts in the absence of treatment (i.e., the $Y(0)$ column of Table 1), by drawing 366 observations from a negative binomial distribution with a mean of 106 and a variance of 402. I chose these parameters to approximate the distribution of non-accidental mortality counts in a large European city. I then defined the counterfactual distribution of mortality by adding the treatment effect, drawn from a Poisson distribution (i.e., the $Y(1)$ column of Table 1). I chose its parameter to increase the number of deaths by 1 on average.[2]

Table 1: Science Table of the Experiment

Day Index	$Y_i(0)$	$Y_i(1)$	τ_i	T_i	Y_i^{obs}
1	122	124	+2	1	124
2	94	96	+2	1	96
3	96	98	+2	0	96
⋮	⋮	⋮	⋮	⋮	
364	96	97	+1	0	96
365	98	98	+0	0	98
366	143	144	+1	1	144

Notes: This table displays the potential outcomes, the unit-level treatment effect, the treatment status and the observed daily number of non-accidental deaths for 6 of the 366 daily units in the scientist's experiment.

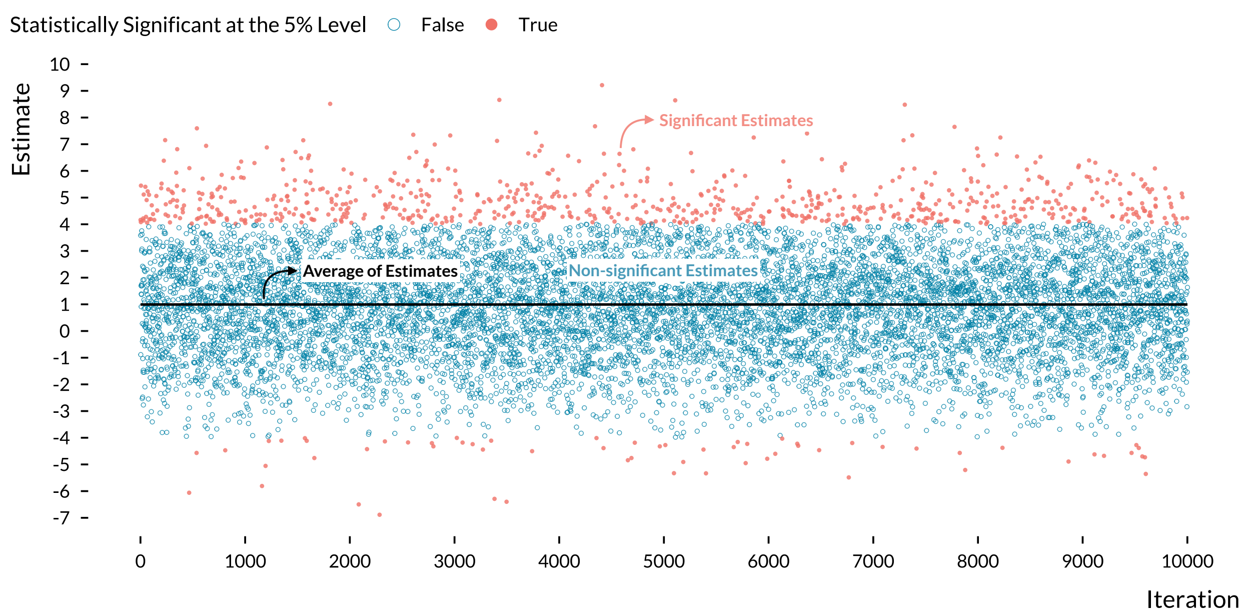
Following the fundamental problem of causal inference, the daily count of deaths the scientist observes is given by the equation: $Y_i^{obs} = T_i \times Y_i(1) + (1 - T_i) \times Y_i(0)$. Considering that the assignment of the treatment was random, how can the statistically significant estimate found by the scientist be 4 times larger than the true treatment effect size? Replicating the experiment a large number of times explains this apparently puzzling result.

[2] In relative terms, the treatment effect size I set represents a 1% increase in the health outcome. The magnitude of this hypothetical effect is larger than the one found in a recent and large-scale study based on 652 cities. Liu et al. (2019) estimated that a 10 $\mu g/m^3$ increase in $PM_{2.5}$ concentration was associated with a 0.68% (95% CI, 0.59 to 0.77) relative increase in daily all-cause mortality.

1.2 Defining Statistical Power and Exaggeration

Figure 1 plots the estimates of 10,000 iterations of the experiment. Even if there is a large variation in the effect size of estimates, their average is equal to the true treatment effect of 1 additional death. We can however note that estimates close to the true effect would not be statistically significant at the 5% level. In a world without publication bias, several replications of this experiment would recover the true treatment effect, on average. Yet, despite recent changes in scientific practices and editorial policies, non-statistically significant estimates are less valued and replication exercises remain rare (Brodeur, Cook and Heyes 2020). In a world with publication bias, statistically significant estimates are more likely to be made public. Out of the 10,000 simulation estimates, about 800 are statistically significant at the 5% level. The *statistical power* of the experiment, which is the probability of rejecting the null hypothesis when there is actually an effect, is estimated to be 8%. The scientist was therefore lucky to get a statistically significant estimate.

Figure 1: 10,000 Replications of the Scientist's Experiment



Notes: Each dot represents a point estimate of one of the 10,000 iterations of the randomized experiment run by the mad scientist. Red dots are statistically significant at the 5% level while blue dots are not. The black solid line represents the average of estimates, equal to the true average effect of 1 additional death.

With such a low statistical power, statistically significant estimates are however not informative of the treatment of interest. Two metrics, the average Type M (magnitude) error and the probability of making a Type S (sign) error, help assess the negative consequences of a lack of statistical power. The exaggeration ratio, or expected Type M error, is defined as the ratio of the absolute values of the statistically significant estimates over the true effect size (Gelman and Carlin 2014). In the present case, with an estimated statistical power of 8%, the scientist could expect their statistically significant estimates to be inflated on average by a factor of 5. We also notice in Figure 1 that a non-negligible fraction of statistically significant estimates are of the wrong sign: this proportion is the probability of making a Type S error (Gelman and Tuerlinckx 2000). In this experiment, a statistically significant estimate has an 8% probability of having the wrong sign.

Formally, the statistical power of a test is the probability of rejecting the null hypothesis $H_0 : \beta = 0$, where β is the true effect of the estimand of interest. For $\hat{\beta}$, a normally distributed unbiased estimator of β with standard error σ, the power of the null hypothesis test at the 5% level is equal to $\Phi(-1.96 - \frac{\beta}{\sigma}) + 1 - \Phi(1.96 - \frac{\beta}{\sigma})$, where Φ is the cumulative distribution function of the standard normal distribution. It increases with β, the true value of the effect and with the precision of the estimator, *i.e.*, when σ decreases. The exaggeration ratio is $\mathbb{E}(\frac{|\hat{\beta}|}{|\beta|} \mid \beta, \sigma, |\hat{\beta}|/\sigma > 1.96)$ and the probability of making a Type S error is given by $\Pr(\frac{\hat{\beta}}{\beta} < 0 \mid \beta, \sigma, |\hat{\beta}|/\sigma > 1.96)$. Zwet and Cator (2021) and Lu, Qiu and Deng (2019) derive closed-form expressions for these quantities. They show that both the exaggeration ratio and the probability of Type S error decrease with β, with the precision of the estimator, and thus with statistical power.

1.3 Exaggeration Risk of Published Estimates in Economics

Exaggeration of published estimates arises from the combination of publication bias and low statistical power. Without publication bias, all results would be published and estimators would be noisy but unbiased on average. Selection on significance is well documented in economics (Rosenthal 1979, Brodeur et al. 2016, Andrews and Kasy 2019, Vivalt 2019, Abadie 2020, Brodeur, Cook and Heyes 2020). Importantly, publication bias extends beyond significance: large or surprising results may also be favored, further amplifying exaggeration when power is limited (Andrews, Kitagawa and McCloskey 2024). Conversely, when statistical power is high, significant estimates are close to the true effect and publication bias has little scope to distort them. Yet, Ioannidis, Stanley and Doucouliagos (2017) estimate the median power across a wide array of economics literatures at 18% and find that nearly 80% of results are exaggerated by a factor of two or more. The two ingredients leading to exaggerated published estimates are thus present in the economics literature. In the following sections, I explore exaggeration and power issues in the literature on the short-term health effects of air pollution and in the broader causal inference literature in economics, and document the potential impacts of exaggeration on the design of public policies.

2 Diagnosing Exaggeration Risk and Policy Implications

From extreme events such as the London Fog of 1952 to the development of sophisticated time-series analyses, a vast epidemiology literature of more than 600 studies has established that air pollution induces adverse health effects over the short term. Increases in the concentration of several ambient air pollutants have been found to be associated with increases in daily mortality and emergency admissions for respiratory and cardiovascular causes (Schwartz 1994, Samet et al. 2000, Le Tertre et al. 2002, Bell, Samet and Dominici 2004, Liu et al. 2019). With this objective in mind, researchers in economics and epidemiology have recently used causal inference methods to improve on the standard epidemiology literature that relied on associations (Dominici and Zigler 2017, Bind 2019). Newly obtained results confirm the short-term health effects of air pollution (Schwartz et al. 2015, Schwartz, Fong and Zanobetti 2018, Deryugina et al. 2019). Based on these results, environmental protection and public health agencies design policies such as air quality alerts to mitigate the burden of air pollution. Accurate estimates of these effects are therefore crucial as they are directly used to implement and update policies to address this major public health issue.

This section first analyzes the literature through the prism of statistical power and exaggeration. It describes how I ran retrospective analyses of the causal inference and standard epidemiology literatures, as well as a broader economics literature. Then, it explores the impacts of exaggeration on the design of concrete public policies through the case of air pollution mitigation policies in the United States.

2.1 Approach to the Literature Analysis

To compute the power, exaggeration ratio, and rate of Type S error of published studies, I rely on retrospective power analyses. The core question these analyses address is: would the design of the study be reliable enough to recover the true effect if it were in fact smaller than the obtained estimate? To do so, one only needs the precision of the estimator, *i.e.*, its standard error, and a hypothesized true effect size. Indeed, the power, exaggeration ratio, and rate of Type S error all depend on the true magnitude of the estimand, which is never observed. Concretely, retrospective power analyses draw many estimates from a hypothetical asymptotic distribution of the estimator, a normal distribution centered on the hypothesized true effect with standard deviation equal to the standard error obtained in the study (Gelman and Carlin 2014). Statistical power is the proportion of sampled estimates that are statistically significant. The exaggeration ratio is the average ratio of the absolute values of significant estimates over the assumed true effect. The Type S error rate is the proportion of significant estimates with the opposite sign of the true value. In my analysis, I use the R package `retrodesign` described in Timm, Gelman and Carlin (2019), which implements closed-form analogues of these simulations (Lu, Qiu and Deng 2019).

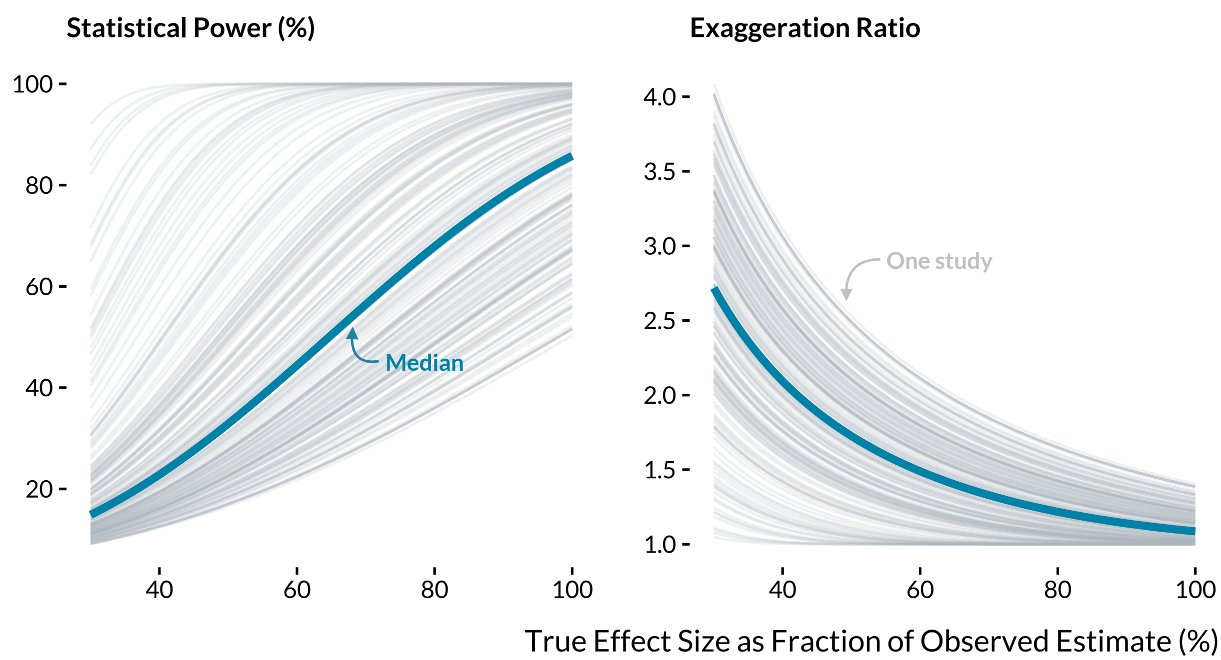
The crux of such analyses lies in the assumptions regarding the magnitude of the hypothesized effect size. In the literature I consider, since treatments and outcomes vary across studies, it is not possible to make general aggregated assumptions on true effect sizes; we need study-specific hypothesized values. I do not claim that any hypothesized effect size is the true effect size but rather ask whether the design of the study would allow accurately capturing an effect of this magnitude. A well-designed study should be able to detect a range of plausible effect sizes, including smaller effect sizes than the obtained estimate and not merely this obtained estimate itself. A design able to accurately recover a small effect will also recover a larger one, but the reverse will not hold. For instance, we would expect a Two-Stage Least-Squares (2SLS) design to have enough power to detect an effect of the magnitude of the corresponding OLS estimate, despite the potential bias of this naive estimate. Similarly, we may want the design to be able to accurately detect effects that are a fraction of the obtained estimate. Using a fraction of the obtained estimate as a hypothetical effect size provides some degree of comparability across otherwise non-comparable studies. I do not claim that this hypothetical effect size is actually the true effect size. The resulting figures should therefore not be read as informative of the actual exaggeration of a given study, but as a way to gauge the strength of its design and to build comparisons across studies. The approach based on fractions of the estimate has important limitations, as hypothesized values are constructed from potentially exaggerated published estimates (Gelman and Carlin 2014). To complement and strengthen the analysis, for a more homogeneous subset of studies, I use meta-analysis estimates from Shah et al. (2015) as alternative true effect sizes.

2.2 Exaggeration Risk in the Causal Inference Literature

Through a thorough search strategy on Google Scholar, IDEAS, and PubMed, I retrieve all studies that (i) focus on the short-term health effects of air pollution on mortality and morbidity outcomes, and (ii) rely on a causal inference method. Appendix A discusses the selection process and displays the list of the 36 articles that match the search criteria. For each study, the method, the health outcome and air pollutant considered, as well as the point estimate and standard error of the main specification, are manually retrieved.

To evaluate potential statistical power issues in this literature, I implement retrospective power calculations, considering a variety of hypothetical true effect sizes. Figure 2 plots the power and exaggeration curves for 186 specifications whose results are statistically significant at the 5% level. It shows that many studies in this literature would not be adequately powered to detect effects meaningfully smaller than the ones they find, though with wide heterogeneity across studies. For a hypothesized effect size equal to half of the obtained estimate, consistent with the typical exaggeration of two found in Ioannidis, Stanley and Doucouliagos (2017) and Ferraro and Shukla (2020), the median power would be 33% and the median exaggeration ratio would be 1.7, with only 11% of studies having a power greater than 80%. One quarter of studies would, on average, exaggerate this hypothetical effect size by a factor greater than 2. High power studies would on the contrary capture such hypothetical effects without any exaggeration. This pattern may help explain the very large effect sizes sometimes observed in the causal inference literature.

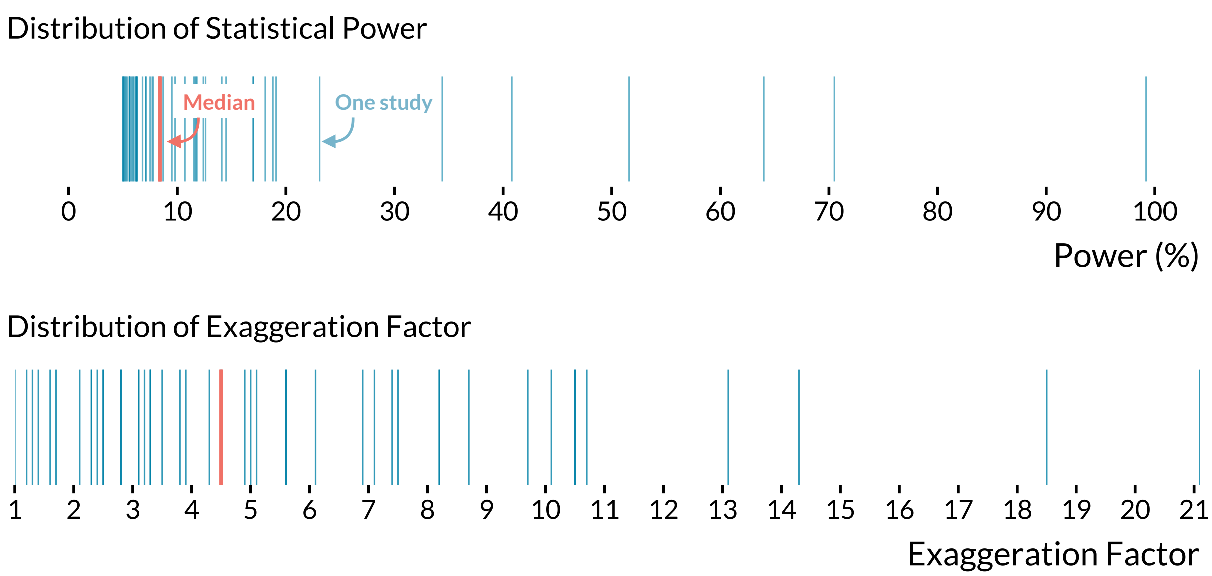
Figure 2: Statistical Power and Exaggeration Curves of Causal Inference Studies



Notes: Each gray line is a power curve or an exaggeration curve of a statistically significant result published in the causal inference literature. The blue lines are the median values. For visual clarity, results with exaggeration ratios that were too large are dropped.

Then, for the 49 instrumental variable results that are statistically significant and that report the corresponding naive regression results, I evaluate whether the 2SLS designs would recover an effect close to that of the naive estimate. I do not claim that the naive estimate is the true effect but use it as a benchmark that a well-powered 2SLS design should be able to recover. Figure 3 displays the distribution of the resulting statistical power and the average exaggeration ratio under this OLS benchmark. The median estimated power is 8.4%, which translates into very large exaggeration ratios: half of the studies would exaggerate a true effect of the size of the OLS estimate by a factor of at least 4.5. For a quarter of studies, the exaggeration ratio exceeds 8.2. This heterogeneity is both reassuring and concerning: while some designs appear adequately powered, a substantial share of studies might suffer from severe exaggeration. Such inflation of statistically significant estimates could explain part of the gap between the standard and causal literatures. This discrepancy could also partly reflect omitted variable bias, attenuation from measurement error, or differences in the causal estimands targeted by each strategy. These explanations are not mutually exclusive, and the lack of power and inability of instrumental variables to recover smaller effect sizes remain concerning.

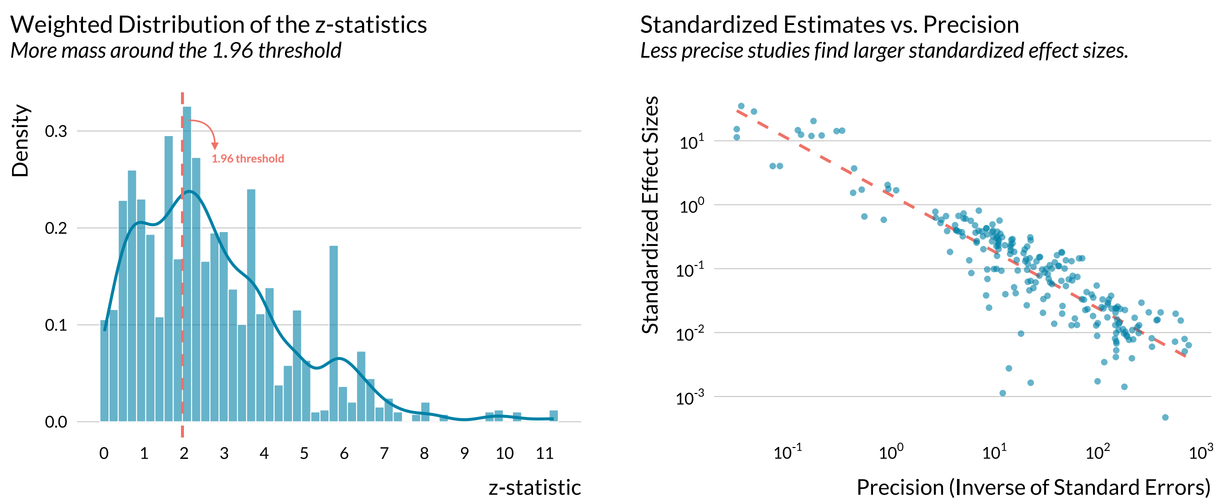
Figure 3: Distribution of Power and Exaggeration Ratio for Instrumental Variable Designs, Benchmarked Against their Corresponding Naive OLS Estimates



Notes: For 49 statistically significant 2SLS estimates, the true values of effect size are defined as the corresponding OLS estimates. Each blue line represents either the estimated statistical power (%) or exaggeration factor of a study's result. Orange lines are the median of the two metrics. For visual clarity, three extreme exaggeration ratios are not displayed.

Exaggeration of significant estimates is problematic when such results are favored for publication. I find suggestive evidence of selection on significance in the causal literature. A caliper test following Brodeur, Cook and Heyes (2020) reveals an excess mass in the weighted z-statistics distribution above the 5% threshold (left panel of Figure 4), and the method of Andrews and Kasy (2019) points in the same direction. The relatively small number of estimates, however, limits my ability to implement more detailed heterogeneity analyses. The right panel of Figure 4 produces further evidence: imprecise designs yield larger standardized effect sizes. This pattern is consistent with exaggeration driving part of the observed variation in effect sizes across studies. If published estimates captured true effects, their standardized magnitude should be independent of precision. The inverse relationship observed in Figure 4 suggests instead that less precise designs yield larger published effects. Combined with the evidence of bunching around significance thresholds in the left panel, these are the two ingredients for exaggerated published estimates. Causal estimates in this literature are typically an order of magnitude larger than epidemiological ones, and the patterns documented here suggest that exaggeration is a plausible contributor to this gap.

Figure 4: Suggestive Evidence of Publication Bias and Exaggeration Risk in the Causal Literature



Notes: The sample in the left panel includes all 382 estimates reported in articles from the causal literature, excluding "naive" OLS estimates and placebo tests. Following Brodeur, Cook and Heyes (2020), the weights are equal to the inverse of the number of tests in the article. In the right panel I exclude the "naive" OLS estimates and placebo tests. Both axes are on a log10 scale. Limiting the sample to economics journals leaves the figures essentially unchanged (see supplemental material). Distinguishing between top 5 and other journals shows that even if their standardized effect sizes are typically smaller in top 5 journals, the same inverse relationship can be observed.

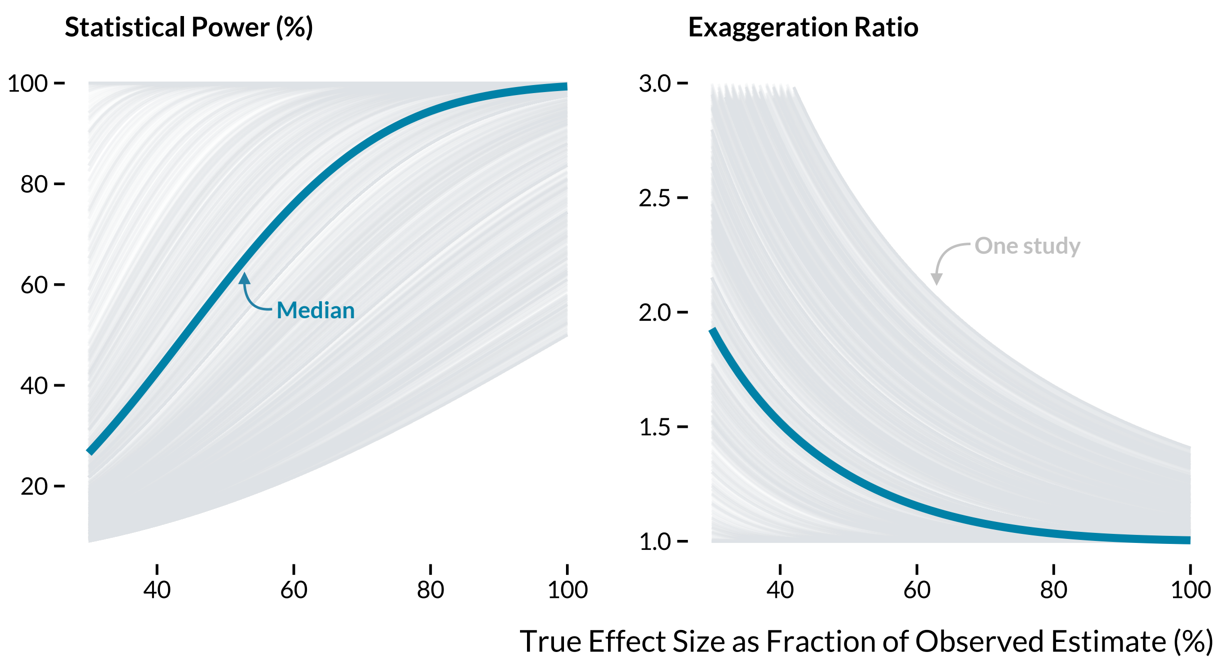
2.3 Exaggeration Risk in the Standard Epidemiology Literature

Hundreds of papers have been published on the short-term health effects of air pollution in epidemiology, medicine, and public health journals. A large fraction of these articles rely on Poisson generalized additive models, which allow for flexible adjustment for the temporal trend of health outcomes and for non-linear effects of weather parameters. This literature spans over 20 years and has replicated analyses in a large number of settings, providing crucial insights into the acute health effects of air pollution.

To gather a corpus of relevant articles, I run a search query on PubMed and Scopus combining air pollution terms with health outcomes, restricting to short-term effects. I retrieve point estimates and confidence intervals from the abstracts of 668 epidemiology articles (2,155 estimates), exploiting the fact that this literature routinely reports these quantities in standardized formats. Appendix A and the project's website describe in detail how these articles were selected and retrieved.

Figure 5 displays the power and exaggeration curves for the 1,982 statistically significant estimates. For instance, 58% of studies would have a power below the conventional 80% threshold to detect an effect size equal to half of the obtained estimate. The median exaggeration ratio would be 1.3, though this masks substantial heterogeneity: for one quarter of studies, the exaggeration ratio would exceed 1.9. This heterogeneity does not seem to be systematically correlated with observable factors, even though health science journals appear less prone to power issues than other journals and the few studies that consider air quality indexes as a pollutant seem to have higher power. Notably, the share of low-power studies has stagnated since the 2000s.

Figure 5: Power and Exaggeration Curves for the Epidemiology Literature

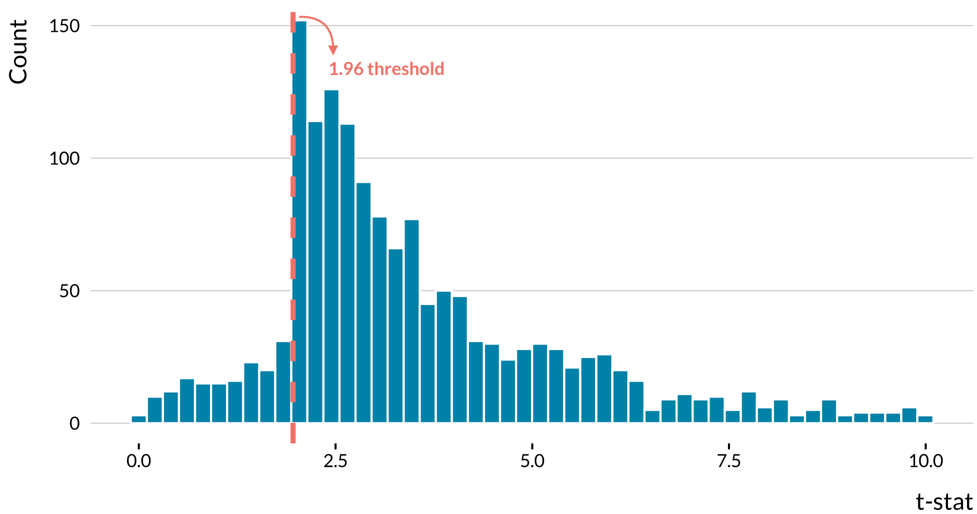


Notes: Each gray line is a power curve or an exaggeration curve of a statistically significant result published in the epidemiology literature. The blue lines are the median values. For visual clarity, results with exaggeration ratios that were too large are dropped.

To complement this analysis, I use meta-analysis estimates from Shah et al. (2015) as better-grounded hypothetical true effect sizes for a subset of 94 studies on the effects of several air pollutants on mortality and emergency admissions for stroke. For each of these studies, I run retrospective power calculations to evaluate their ability to retrieve the corresponding meta-analysis estimate. I find that 63% have a power below 80%, and the median exaggeration ratio is 1.6. As shown in Figure B.1, results vary considerably by pollutant, from a median exaggeration of 1 for $PM_{2.5}$ to 13.4 for O_3.

Unlike the causal inference corpus, in this epidemiology corpus, I do not observe the full set of published estimates but only those reported in those studies' abstracts. These are headline results that authors chose to emphasize and that are most likely to inform policy-making. As shown in Figure 6, there is striking bunching at the 1.96 threshold, indicating that statistical significance strongly influences which results are highlighted, constituting a specific form of selection on significance. Since identification approaches are relatively uniform across this literature, studies are less likely to be evaluated on methodological grounds, making the significance and magnitude of results all the more central to publication decisions.

Figure 6: Distribution of t-statistics of Estimates in Abstracts from the Standard Epidemiology Literature



Notes: All estimates come from abstracts. Estimates with t-stat > 10 are not reported here for readability. The vertical dashed line represents the usual 1.96 threshold.

2.4 Exaggeration in the Broader Causal Economics Literature

The patterns documented above are not specific to the air pollution literature. I leverage two existing datasets to document them. First, using data from Brodeur, Cook and Heyes (2020), which reviews the universe of hypothesis tests reported in papers published in the top-25 economics journals in 2015 and 2018 using RCT, DID, RDD, or IV, I implement retrospective power calculations following the same approach as above. Considering a hypothesized true effect equal to half of the obtained estimate, the median power and exaggeration ratio would be 37% and 1.6, respectively. Only 22% of designs would exceed the 80% power threshold, while 28% would exaggerate true effects by more than a factor of 2 on average. Results are relatively comparable across methods.

Second, I focus on IV designs using data from Young (2022), covering 30 reproducible papers published in *American Economic Association* journals.[3] The median ratio of headline IV to corresponding OLS estimates is 2.3, with 25% of estimates exceeding a ratio of 5.4. IV designs of significant headline results would have a median power of only 25% to detect an effect of the magnitude of the OLS estimate. Significant estimates from half of these designs would exaggerate the OLS estimate by a factor larger than 2, and a quarter by a factor larger than 3.5. Only 17% would reach the conventional 80% power threshold.

The heterogeneity and importance of exaggeration documented in the case of studies on the short-term health effects of air pollution thus extend to broader economics literatures.

2.5 Impacts of Exaggeration on Public Policy

Academic estimates of the short-term health effects of air pollution directly inform environmental policy. Consider the case of the United States. Under the Clean Air Act, the Environmental Protection Agency (EPA) uses systematic reviews of the scientific literature, synthesized in Integrated Science Assessments (ISAs), to set National Ambient Air Quality Standards (NAAQS) and conducts Regulatory Impact Analyses (RIAs) comparing the costs and monetized health benefits of alternative regulatory approaches (U.S. EPA 2015; 2024*a*).[4] To quantify health benefits, the EPA relies on the Benefits Mapping and Analysis Program (BenMAP), a modeling tool that combines air quality projections with concentration-response coefficients from the epidemiology literature (Sacks et al. 2018).[5] In this framework, monetized health benefits are computed as $\Delta Y = \text{Pop} \times Y_0 \times (1 - e^{-\hat{\beta} \times \Delta AQ})$ where ΔY denotes the change in health outcomes attributed to air pollution, Pop the exposed population, Y_0 the baseline incidence, ΔAQ the specified change in air quality, and $\hat{\beta}$ the estimated health effect coefficient. For the small concentration changes typically considered in regulatory analysis, this simplifies to $\Delta Y \approx \text{Pop} \times Y_0 \times \hat{\beta} \times \Delta AQ$. These benefits are then valued using measures such as the Value of a Statistical Life, whose EPA central estimate is approximately \$10.7 million in 2022 dollars (U.S. EPA 2024*b*). Because monetized ben-

[3] The exact selection process is described in Section 3 of Young (2022). The sample consists of papers using the keyword "instrument", published through July 2016, that are reproducible, rely on open-access data and Stata code, and use linear methods with standard covariance estimates.

[4] This reflects the regulatory framework in place through the end of 2024.

[5] States are responsible for implementing these standards through State Implementation Plans (SIPs), and often draw on EPA-provided analytical tools such as BenMAP-CE (Community Edition), as well as guidance from RIAs and associated technical documents, to select compliance measures (U.S. EPA 2023).

efits are proportional to $\hat{\beta}$, any exaggeration of the health effect coefficient passes through linearly into the benefit-cost analysis. This is an inherent feature of EPA's quantification framework: if the estimated health effects are exaggerated by a factor $\zeta = \frac{\hat{\beta}}{\beta}$, perceived benefits will be exaggerated by the same factor.

To illustrate how exaggeration could affect benefit-cost analyses, I apply the range of exaggeration ratios documented above to the marginal benefits computed in the 2024 RIA. The resulting estimates should be read as illustrative scenarios rather than direct estimates of the distortion in any specific policy cost-benefit analysis. In addition, there is growing interest in replacing estimates based on association with causal ones (Dominici and Zigler 2017, Bind 2019). I examine what would happen if a specific causal estimate were used instead of a standard epidemiology estimate. For instance, one of the studies used in EPA's syntheses of the literature (ISAs) exploring the short-term effect of $PM_{2.5}$ on mortality, Zanobetti and Schwartz (2009), finds that a 10 μg/m^3 increase in $PM_{2.5}$ concentration is associated with a 0.98% (s.e. = 0.12) increase in all-cause mortality. Among the causal inference studies in the corpus considered above, Schwartz et al. (2015) estimate the same relationship using an instrumental variable approach and find a 5.3% (s.e. = 2.24) effect, roughly five times the estimate found in Zanobetti and Schwartz (2009). Taking Zanobetti and Schwartz (2009) as a plausible true effect size, the Schwartz et al. (2015) design would have a statistical power of 7.2% and an expected exaggeration ratio of 5.4.

To explore concrete implications of exaggeration for policy, consider the 2024 RIA for the revision of the $PM_{2.5}$ standards, which lowered the annual standard from 12 to 9 μg/m^3 (U.S. EPA 2024*a*). The RIA estimates monetized benefits in 2032 between \$22 billion and \$46 billion (2017\$) against annual costs of \$590 million. The RIA also evaluates alternative standard levels, at 10 μg/m^3, estimating benefits of \$8.5 to \$17 billion against costs of \$200 million. The marginal benefits of tightening from 10 to 9 μg/m^3 are thus approximately \$13.5 to \$29 billion, against marginal costs of \$390 million. Table 2 describes the impact of various potential exaggeration ratios on the expected marginal net benefits.

Under the potential exaggeration implied by the comparison between Schwartz et al. (2015) and Zanobetti and Schwartz (2009), marginal benefits would fall to approximately \$2.5 to \$5.4 billion against marginal costs of \$390 million, substantially weakening the case for the most stringent standard. The marginal benefits to cost ratio of the policy would fall from 34 to 74 to 6 to 14. Air pollution regulations have benefit-cost ratios large enough that even substantial exaggeration leaves net benefits positive. In domains where estimated benefits and costs are closer in magnitude, exaggeration could reverse the sign of a benefit-cost analysis and therefore determine whether a regulation is implemented. Moreover, my analysis using meta-analysis benchmarks from Shah et al. (2015) reveals that exaggeration varies substantially across pollutants (Figure B.1). Exaggeration could therefore differen-

Table 2: Policy Implications of Documented Exaggeration Risk

	Exaggeration ratio	Expected Marginal Benefits (B$/year)	Adjusted Marginal Benefits (B$/year)	Marginal Benefits to Costs Ratio
Epidemiology median	1.3	13.5 to 29	10.4 to 22.3	27 to 57
Epidemiology 25th pctile	1.9	13.5 to 29	7.1 to 15.3	18 to 39
Causal inference median	1.7	13.5 to 29	7.9 to 17.1	20 to 44
Causal inference 25th pctile	2.0	13.5 to 29	6.8 to 14.5	17 to 37
IV vs OLS median	4.5	13.5 to 29	3 to 6.4	8 to 17
Causal vs standard example	5.4	13.5 to 29	2.5 to 5.4	6 to 14

Notes: Adjusted marginal benefits represent the benefits under the hypothesis that health effects were exaggerated: due to the linearity of the relationship between health effects and monetized benefits, they are simply computed as the expected benefits of tightening the policy from 10 to 9 $\mu g/m^3$ divided by the exaggeration ratios found in our analysis of the literature. The various exaggeration ratios considered are drawn from the analysis of the literature above, except for the last one which comes from the comparison between Schwartz et al. (2015) and Zanobetti and Schwartz (2009).

tially affect the benefit-cost analyses of pollutants. If, for instance, it inflates monetized benefits by a factor of 3 for $PM_{2.5}$ and 5 for O_3, ozone regulation could appear more attractive relative to $PM_{2.5}$ regulation than it actually is, potentially even reversing the ranking between the two.

3 Approach for the Simulation Analysis

While the analysis of the literature can provide evidence of exaggeration and lack of statistical power, it does not allow for clearly identifying or establishing the parameters driving it. I therefore implement a prospective analysis to overcome this limitation (Altoè et al. 2020, Black et al. 2022, for additional guidance on power calculations in complex observational settings). I run Monte Carlo simulations based on real data to emulate the main empirical strategies found in the literature. I use real data to avoid the difficult task of modeling the long-term and seasonal variations in health outcomes as well as the specific effects of weather variables such as temperature. These simulations are grounded in the literature on the acute health effects of air pollution but nonetheless share many similarities with prevailing applied economics settings: panel data, large data sets, canonical identification strategies.

The present section describes how the simulations are implemented. But before that, I present the causal identification strategies used to measure the acute health effects of air pollution and provide a brief description of the data used for the simulations.

3.1 Typical Research Designs to Measure the Short-Term Health Effects of Air Pollution

Several empirical strategies have been leveraged to estimate the short-term health effects of air pollution. I simulate the main ones existing in the literature. I consider the usual setting where data on air pollution, weather parameters, and health outcomes are at the daily-city level.

Standard regression approach. The standard strategy consists in directly estimating the dose-response between an air pollutant and a health outcome. In the epidemiology literature, researchers often rely on Poisson generalized additive models where they regress the daily count of a health outcome on an air pollutant concentration, while flexibly adjusting for weather parameters, seasonal and long-term variations. I approximate the workhorse model used by epidemiologists using linear models estimated via ordinary least squares:

$$\mathrm{Y}_{c,t} = \alpha + \beta \mathrm{P}_{c,t} + \mathbf{W}_{c,t}\lambda + \mathbf{C}_t\gamma + \epsilon_{c,t} \tag{1}$$

where c is the city index and t the daily time index. $\mathrm{Y}_{c,t}$ is the daily count of cases of a health outcome and $\mathrm{P}_{c,t}$ the average daily concentration of an air pollutant and $\epsilon_{c,t}$ an error term. The parameter β captures the short-term effect of an increase in the air pollutant concentration on the health outcome. To address confounding issues, the model adjusts for a set of weather covariates, $\mathbf{W}_{c,t}$, and calendar indicators $\mathbf{C}_t$.

Instrumental variable (IV) approach. The standard strategy could be prone to omitted variable bias and measurement error. A growing number of articles therefore rely on exogenous variations in air pollution. Most causal inference papers rely on IV designs where the concentration of an air pollutant is instrumented by thermal inversions (Arceo, Hanna and Oliva 2016), wind patterns (Schwartz, Fong and Zanobetti 2018, Deryugina et al. 2019), extreme natural events such as sandstorms or volcanic eruptions (Ebenstein, Frank and Reingewertz 2015, Halliday, Lynham and de Paula 2019), or variations in transport traffic (Moretti and Neidell 2011, Knittel, Miller and Sanders 2016, Schlenker and Walker 2016). This approach can be summarized with a two-stage model where the first stage is:

$$\mathrm{P}_{c,t} = \delta + \theta \mathrm{Z}_{c,t} + \mathbf{W}_{c,t}\eta + \mathbf{C}_t\kappa + e_{c,t} \tag{2}$$

where $\mathrm{Z}_{c,t}$ is the instrumental variable. The second stage is then:

$$\mathrm{Y}_{c,t} = \alpha + \beta \widehat{\mathrm{P}}_{c,t} + \mathbf{W}_{c,t}\lambda + \mathbf{C}_t\gamma + \epsilon_{c,t} \tag{3}$$

where $\widehat{P}_{c,t}$ is the exogenous variation in an air pollutant predicted by the instrument. The causal effect measured by this approach is a weighted average of per-unit causal responses to an increase in the concentration of an air pollutant (Angrist and Imbens 1995).

Reduced-form approach. A subset of articles directly estimates the relationship between the health outcome and exogenous shocks to air pollution. For instance, articles using this approach exploit public transport strikes or thermal inversions as exogenous shocks (Bauernschuster, Hener and Rainer 2017, Jans, Johansson and Nilsson 2018, Godzinski, Castillo et al. 2019, Giaccherini, Kopinska and Palma 2021). They estimate a model of the form:

$$Y_{c,t} = \alpha + \beta D_{c,t} + \mathbf{W}_{c,t}\lambda + \mathbf{C}_t\gamma + \epsilon_{c,t} \tag{4}$$

where $D_{c,t}$ is a dummy equal to 1 when city c is affected by a shock at time t and 0 otherwise. The parameter β captures an intention-to-treat effect.

Regression discontinuity design (RDD) approach. Another empirical strategy found in the literature measures the effects of air quality alerts with a regression discontinuity design (Chen et al. 2018, Anderson, Hyun and Lee 2022). In this approach, the following model is estimated for observations within an air pollution concentration bandwidth around the alert threshold:

$$Y_{c,t} = \alpha + \beta \mathbf{1}\{P_{c,t} > P_c^{(a)}\} + \mathbf{W}_{c,t}\lambda + \mathbf{C}_t\gamma + \epsilon_{c,t} \tag{5}$$

where $P_c^{(a)}$ is the air pollution alert threshold for city c. I restrict my simulations to the case of sharp RDD. This model estimates the intention-to-treat effect of air quality alerts. It can capture the effect of a subsequent decrease in air pollution caused both by traffic restriction policies and inhabitants' avoidance behavior.

3.2 Data

The simulation exercises rely on a subset of the US National Morbidity, Mortality, and Air Pollution Study (NMMAPS). The dataset is publicly available and has been used in several major studies in the early 2000s to measure the short-term effects of ambient air pollutants on mortality outcomes (Peng and Dominici 2008). Specifically, I leverage data at the city-day level for 68 cities over the 1987-1997 period. It corresponds to 4,018 daily observations per city, for a total sample size of 273,224 observations. It contains observations on the average temperature (°C), the standardized concentration of carbon monoxide (CO), and mortality counts for several causes. I focus on CO as it is the air pollutant measured in most cities over the period and its concentration is strongly correlated to that of other pollutants such as particulate matter. Less than 5% of carbon monoxide concentrations and average temperature readings are missing in the initial data set. These missing observations are imputed using the chained random forest algorithm implemented in the `missRanger` package (Mayer 2019).

3.3 Simulations Set-Up

General procedure. My simulation procedure follows 7 main steps:

1. Randomly draw a study period and a sample of cities.
2. For instrumental variable, reduced-form and regression discontinuity designs, randomly allocate days to exogenous shocks/air quality alerts.
3. Modify the health outcome, adding a treatment effect to recover.
4. Estimate the model.
5. Store the point estimate of interest and its standard error.
6. Repeat the procedure 1,000 times.
7. Compute the proportion of statistically significant estimates at the 5% level (the power), the average of the absolute value of significant estimates over the true effect size (the exaggeration ratio), and the proportion of significant estimates of the opposite sign of the true effect (the probability of making a Type S error).

Modeling assumptions. To only capture the specific issues arising due to low statistical power, I build my simulations such that (i) they meet all the required assumptions of empirical strategies and (ii) make it easier, compared to real settings, to recover the treatment effect. For all research designs, the treatment added to the data is not biased by unmeasured confounders or measurement errors. For instrumental variable and reduced-form strategies, I only simulate binary and randomly allocated exogenous shocks (*e.g.*, the occurrence of a thermal inversion), abstracting from concerns such as endogenous selection into instrument exposure. For the regression discontinuity approach, I only model sharp designs where an air quality alert is always activated above a randomly chosen threshold. The simulations always retrieve the true value of the treatment effect on average. Given these conservative assumptions, the simulations can be read as describing the impact of general design parameters rather than of identification-specific concerns. The exaggeration they document is conservative compared to what would arise in more realistic, and noisier, settings. The main goal of these simulations is to describe drivers rather than to quantify exaggeration.

Two approaches for simulating research designs. For the reduced-form and regression discontinuity designs, I follow the Neyman-Rubin causal framework by simulating all potential outcomes (Rubin 1974). Consider that the health outcome value recorded in the NMMAPS dataset corresponds to the potential outcome $Y_{c,t}(0)$. To create the counterfactuals $Y_{c,t}(1)$, I add a treatment effect drawn from a Poisson distribution whose parameter corresponds to the magnitude of the treatment. I then randomly draw the treatment indicators $T_{c,t}$ for exogenous shocks or air quality alerts. For reduced-form strategies, the treatment status of each day is drawn from a Bernoulli distribution with parameter equal to the proportion of exogenous shocks desired. For air pollution alerts, I randomly draw a threshold from a uniform distribution and select a bandwidth such that it yields the desired proportion of treated observations. I finally express the observed values Y^{obs} of potential outcomes according to the treatment assignment: $Y^{obs}_{c,t}$ = (1-$T_{c,t}$)×$Y_{c,t}(0)$ + $T_{c,t}$×$Y_{c,t}(1)$.

To simulate standard regression and the instrumental variable strategies, I rely on a model-based approach. For the standard regression strategy, I first estimate the following statistical model on the data:

$$Y_{c,t} = \alpha + \beta P_{c,t} + \mathbf{W}_{c,t}\lambda + \mathbf{C}_t\gamma + \epsilon_{c,t} \tag{6}$$

I then predict new observations of $Y_{c,t}$ using the estimated coefficients of the model ($\hat{\beta}$, $\hat{\lambda}$, and $\hat{\gamma}$) and by adding noise drawn from a normal distribution with variance equal to that of the residuals $\widehat{\epsilon_{c,t}}$ (Peng, Dominici and Louis 2006). I modify the slope of the dose-response relationship by changing the value of the air pollution coefficient β. For the instrumental variable strategy, I use the same method as for the standard regression approach but first modify observed air pollutant concentrations $P_{c,t}$ according to the desired effect size θ of the randomly allocated instrument:

$$\widetilde{P}_{c,t} = P_{c,t} + \theta Z_{c,t} \tag{7}$$

I draw the allocation of each day to an exogenous shock from a Bernoulli distribution with parameter equal to the proportion of exogenous shocks. I then estimate a two-stage least squares model (2SLS) and modify the coefficient for the effect of the air pollutant on a health outcome. I finally generate the fake observations of the health outcome by combining the prediction from the modified 2SLS model and noise drawn from a normal distribution with variance equal to that of the residuals.

Varying parameters. To understand which parameters affect statistical power issues, I modify one aspect of the research design while keeping other parameters constant. I study the influence of four main parameters. First, I vary the sample size by drawing a different number of cities and changing the length of the study period. Second, I consider different effect sizes of air pollution or of an exogenous shock on the health outcome. Third, I allocate increasing proportions of exogenous shocks/air quality alerts. Fourth, I vary the number of cases in the outcome by considering different health outcomes.

Simulations of Case Studies. The simulations described above help explore the effect of each parameter on statistical power issues. Yet, the resulting set of parameters considered may not be perfectly representative of actual studies. To address this concern, I also calibrate simulation parameters to reproduce three papers published in the literature. I report these analyses in Appendix D.

4 Results of the Simulation Analysis

This section analyzes how statistical power, exaggeration ratio and Type S error are affected by the value of different study parameters. To do so, I set baseline values for these parameters and vary the value of each of them one by one. This provides a sense of the impact of each parameter, other things held equal. The baseline parameters are the following:

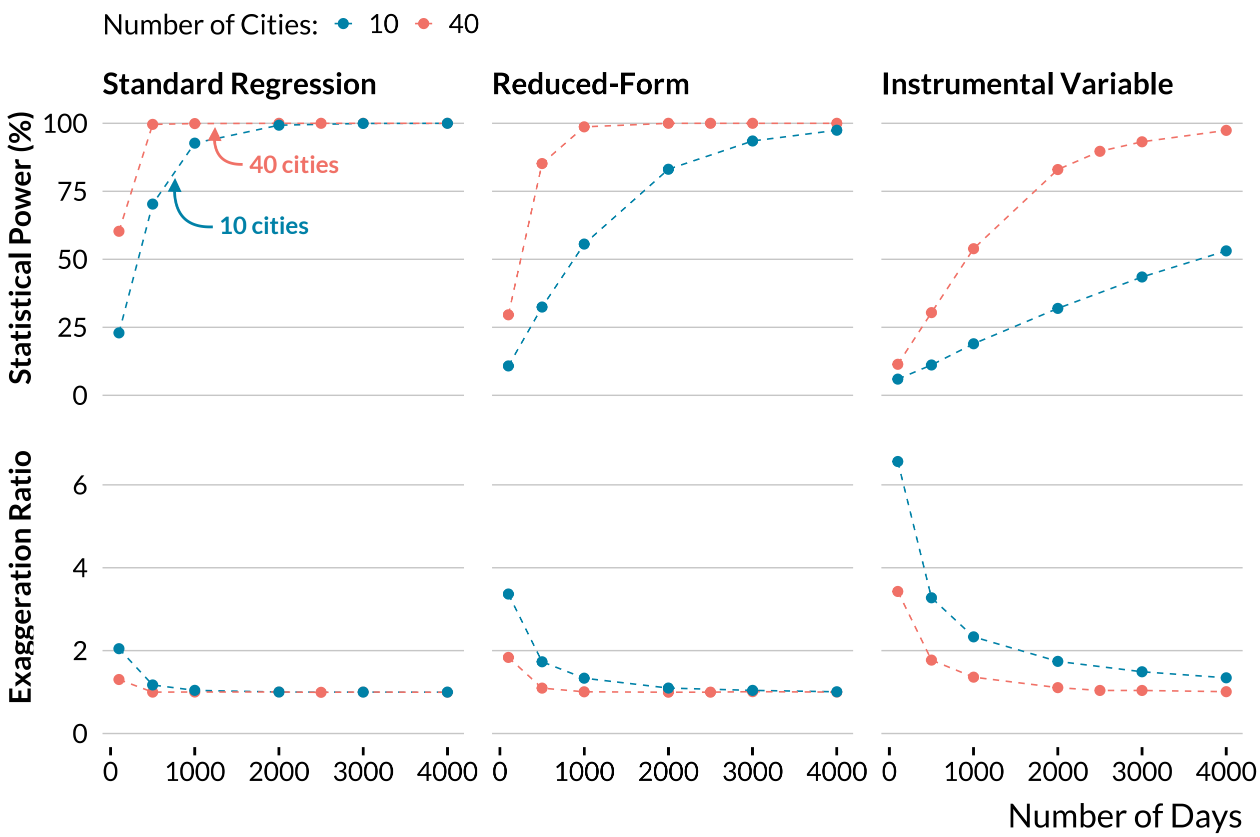
- A large sample size of 100,000 observations (2,500 days × 40 cities),
- A 1% effect size, the order of magnitude found in the most precise studies of the literature. A one standard deviation increase in air pollution or a typical exogenous shock increases the health outcome by 1%,
- 50% of observations are subject to an exogenous shock. For air pollution alerts analyzed with regression discontinuity designs, I only consider observations close to the threshold, resulting in a smaller proportion of treated units: 10%,
- The health outcome is the total daily number of non-accidental deaths. It is the health outcome with the largest average number of counts (average daily mean of 23 cases).

For all statistical models, I adjust for temperature, temperature squared, city and calendar (weekday, month, year, and month×year) fixed effects. I also repeat the simulations for a smaller sample size of 10,000 observations. In Appendix D, I show that statistical power issues can be substantial for actual parameter values found in the literature on acute health effects of air pollution.

4.1 Sample Size

I first find that statistical power and exaggeration issues can arise even for a large number of observations. For a sample size of 40,000 observations, the instrumental variable

Figure 7: Evolution of Power and Exaggeration with Sample Size



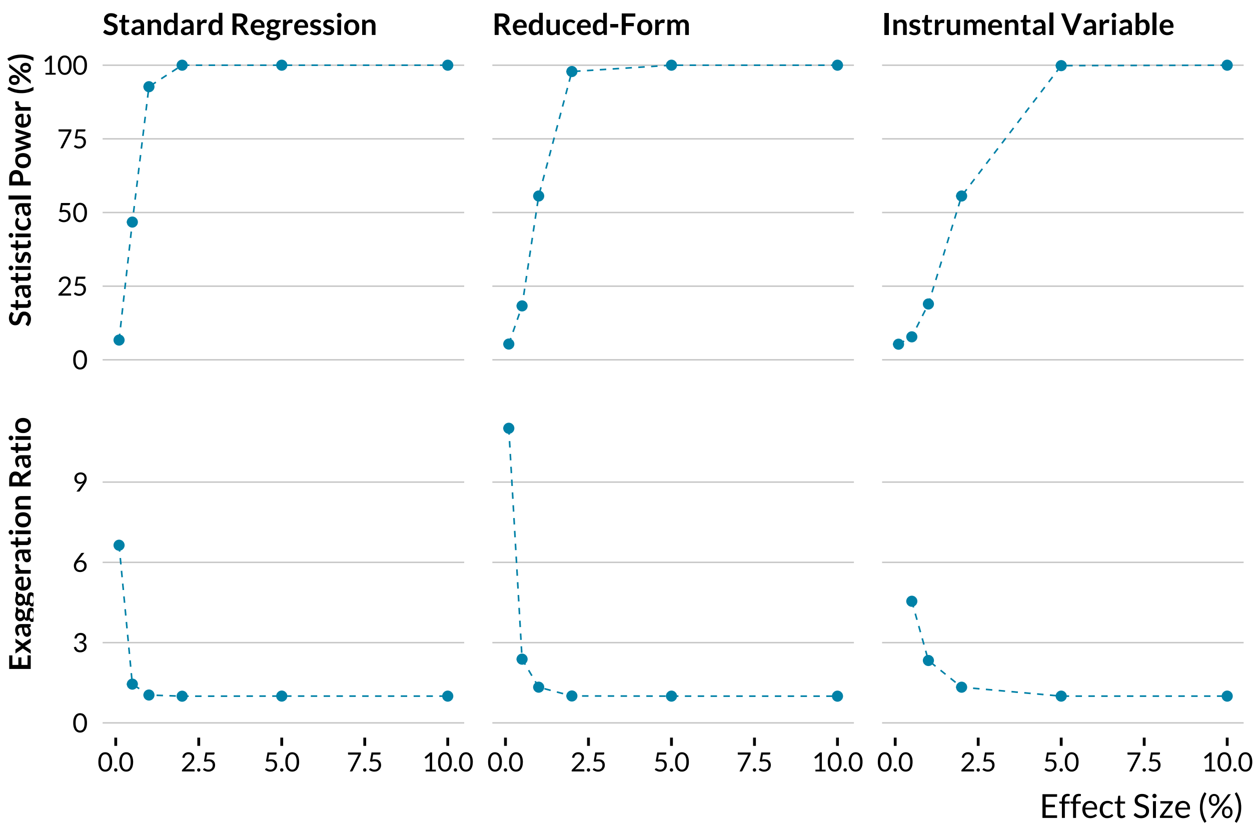
Notes: The other parameters are set to their baseline values: a true effect size of 1%, 50% of observations subject to an exogenous shock for instrumental variable and reduced-form designs, and the health outcome is the total number of non-accidental deaths.

strategy only has a statistical power of 54% and exaggerates the true effect by a factor of 1.4, even with all other parameters set to conservative baseline values. As expected, Figure 7 also shows that the exaggeration ratio decreases and statistical power increases with the number of observations, as both quantities are driven by the variance of the estimator, which decreases with sample size (Zwet and Cator 2021, Lu, Qiu and Deng 2019). The standard regression strategy is much less prone to power issues than the instrumental variable strategy, as the variance of the two-stage least-squares estimator is larger than that of the ordinary least-squares estimator. In the simulations, the probability of making a Type S error is close to zero for all identification strategies and sample sizes.

4.2 Effect Size

Even with the large baseline sample size I consider, statistical power issues appear for effect sizes routinely found in the literature. For instance, for the instrumental variable strategy and an effect size of 0.5%, representative of effect sizes commonly found in the literature,

Figure 8: Evolution of Power and Exaggeration with Effect Size



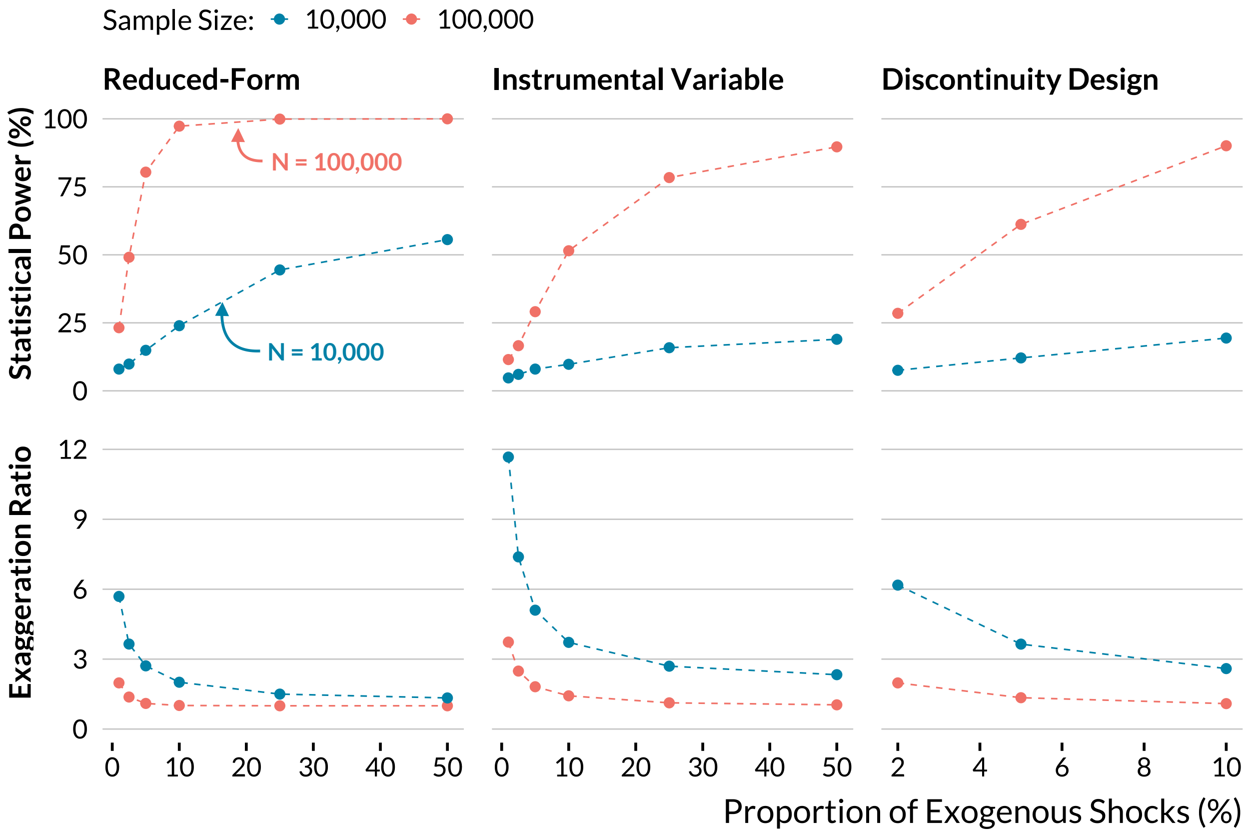
Notes: The sample size is 10,000. The other parameters are set to their baseline values: 50% of observations subject to an exogenous shock for instrumental variable and reduced-form designs, and the health outcome is the total number of non-accidental deaths. For an effect size of 1%, I do not display the exaggeration ratio of the instrumental variable design since it is above 20 and it would distort the graph.

the average exaggeration ratio is about 1.7. As expected, Figure 8 also confirms that the exaggeration ratio decreases and statistical power increases with the true effect size (Zwet and Cator 2021, Lu, Qiu and Deng 2019). Standard regression and reduced-form strategies are less prone to power issues, even for small effect sizes.

4.3 Proportion of Exogenous Shocks

I find that relying on rare exogenous shocks can produce substantial exaggeration, even for large sample sizes and effect sizes. Figure 9 shows that statistical power increases and exaggeration decreases with the proportion of treated units for instrumental variable, regression discontinuity, and reduced-form designs. In practice, air pollution alerts, thermal inversions, or transportation strikes can represent less than 5% of the observations. With a simulated dataset of 10,000 observations, a proportion of exogenous shocks of 5%, and the baseline effect size, the reduced-form strategy yields an average exaggeration ratio of 2.7.

Figure 9: Evolution of Power and Exaggeration with the Proportion of Exogenous Shocks



Notes: The other parameters are set to their baseline values: a true effect size of 1% and the health outcome is the total number of non-accidental deaths. The proportion of exogenous shocks corresponds to the fraction of days in the sample that are allocated to the treatment.

As discussed in Appendix D, in some economics studies the proportion of treated observations is as low as 0.5% and amounts to fewer than 60 treated units. For a proportion of 1% of treated observations, significant estimates are on average 11 times larger than the true effect. The issue persists for larger samples: with 100,000 observations and 1% of shocks, which still amounts to 1,000 treated units, the exaggeration ratio is 3.7.

This result can be explained by the fact that precision is maximized when half of the observations are exposed to the treatment of interest. The variance of the estimator of the average treatment effect (ATE) is $\sigma^2/[n \times p(1-p)]$ where σ is the standard deviation of the outcome in the treated and control groups and p the proportion of treated units. This quantity increases when p departs from 0.5. While routinely discussed for randomized controlled trials, this point is seldom raised in the context of non-experimental studies. Another way to interpret this result is to consider that a small number of exogenous shocks limits the variation that can be leveraged to identify the effect of interest. When the proportion of shocks decreases, the variance of the treatment variable decreases and therefore the variance of the estimator increases. Similar reasoning can be applied to IV strategies. The total sample size can thus give a misleading picture of the precision of a study when only a small fraction of observations are treated.

4.4 Average Count of Cases of the Health Outcome

Table 3: Evolution of Power and Exaggeration with the Average Number of Daily Cases of Health Outcomes

	Non-Accidental	Respiratory	COPD
Number of Cases	23	2	0.3
Statistical Power (%)	90	16	7.5
Exaggeration Ratio	1	2.4	5.9

Notes: This table displays the average number of cases, the power, and the exaggeration ratio for three health outcomes: non-accidental deaths, respiratory deaths, and chronic obstructive pulmonary deaths for individuals aged between 65 and 75. These figures are obtained for the instrumental variable design with a sample size of 100,000 and 50% of observations subject to an exogenous shock. The instrumental variable increases the air pollutant concentration by 0.5 standard deviation. A one standard deviation increase in the instrumented air pollutant leads to a 1% relative increase in the health outcome considered.

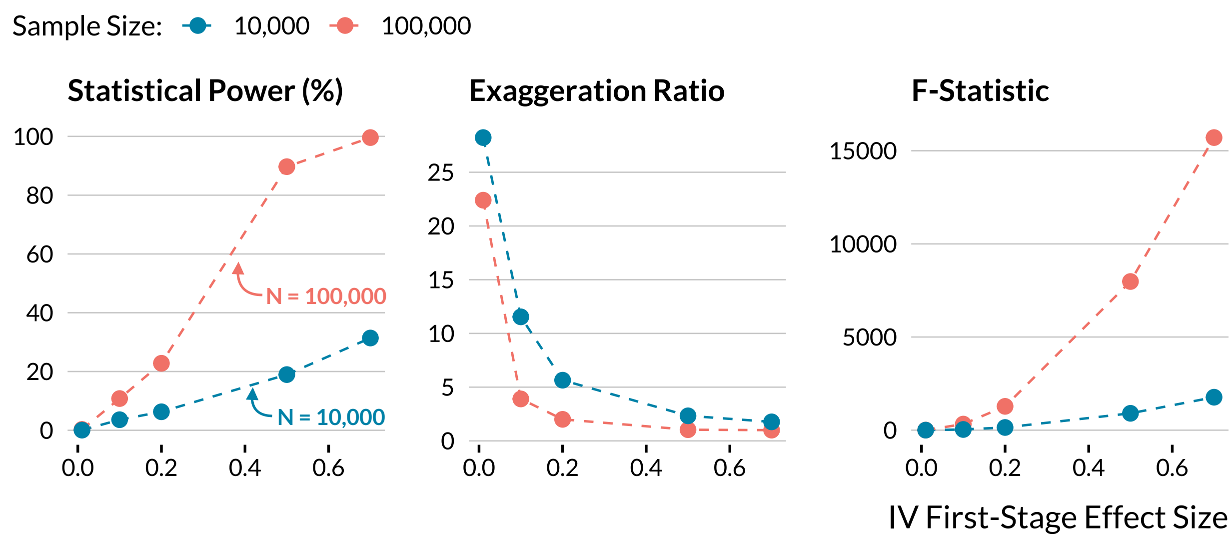
I show in Table 3 that the count of cases of the outcome can critically affect exaggeration. To explore situations with various numbers of cases, I consider three different outcome variables: the total number of non-accidental deaths (daily mean $\simeq$ 23), the total number of respiratory deaths (daily mean $\simeq$ 2) and the number of chronic obstructive pulmonary disease (COPD) cases for individuals aged between 65 and 75 (daily mean $\simeq$ 0.3). Using the baseline parameters with the large dataset, statistical power is close to 100% for a 1% increase in the total number of non-accidental deaths. However, statistical power drops sharply when the average count of cases decreases. The instrumental variable strategy has only 16% statistical power to detect a 1% increase in respiratory deaths, with an average exaggeration ratio of 2.4. For chronic obstructive pulmonary deaths, the exaggeration ratio reaches 5.9. In a setting with only a few deaths per day, a 1% increase in the number of deaths will rarely cause additional deaths, making the effect harder to detect.

Subgroup analyses are routinely carried out in the literature to evaluate the acute health effects of air pollution on children or the elderly. These populations are more vulnerable to air pollution, which can lead to larger effect sizes that attenuate exaggeration concerns. Yet the lower number of cases in these subgroups exacerbates them, creating a trade-off for power issues. This concern applies broadly: any study estimating effects on outcomes with low counts faces the same constraint.

4.5 Strength of the Instrument

I find that substantial exaggeration arises when the instrument only explains a limited portion of the variation in the explanatory variable of interest, even when first-stage F-statistics are large. In my simulations, I consider a binary instrument (e.g., the occurrence of a thermal inversion or a public transport strike) and define its strength as its standardized effect on the air pollutant concentration. As shown in Figure 10, statistical power decreases and exaggeration increases sharply when this strength decreases.

Figure 10: Evolution of Power and Exaggeration with the Strength of the Instrumental Variable



Notes: The true effect size is a 1% relative increase in the health outcome. The health outcome used in the simulations is the total number of non-accidental deaths. Half of the observations are exposed to exogenous shocks. The strength of the instrumental variable is defined as its effect in standard deviation on the air pollutant concentration.

This pattern arises because the variance of the 2SLS estimator depends on the correlation between the instrument and the endogenous variable. In the homoskedastic case, the asymptotic variance of the 2SLS estimator is $\left(\mathbb{E}[XZ']\mathbb{E}[ZZ']^{-1}\mathbb{E}[ZX']\right)^{-1}\sigma^2$, where σ^2 is the variance of the error, X the endogenous variable, and Z the instrument. When $\mathbb{E}[XZ']$ and $\mathbb{E}[ZX']$ decrease, the variance of the estimator increases, leading to low power and exaggeration. Yet a small correlation does not necessarily translate into a small first-stage F-statistic. This F-statistic can be decomposed as $F = n \cdot R^2/(1-R^2)$ (Keane and Neal 2024): it increases with both the R^2 of the first stage and the sample size n. A large sample can thus produce a large F-statistic even when the instrument only explains a small share of the variation in the endogenous variable. With 100,000 observations, an instrument strength of 0.2, and an effect size of 1%, the average F-statistic is 1,278, yet statistical power is only 23% and the average exaggeration ratio is 2. The F-statistic indicates that the first-stage coefficient is precisely estimated, but not that it is large enough for the second-stage estimator to be well powered. A large F-statistic could therefore hide large exaggeration issues: a design with an F-statistic well above conventional thresholds can still suffer from substantial exaggeration when the instrument explains only a limited share of the variation in the endogenous variable.

In practice, the drivers highlighted in this section rarely appear in isolation. A study exploiting rare exogenous shocks in a vulnerable subpopulation combines a small proportion of treated observations with a low outcome count, compounding the power loss and exaggeration risk from each. The case studies in Appendix D illustrate how such combinations can arise in realistic designs.

5 Discussion

Growing evidence documents statistical power issues and publication bias in economics, causing exaggeration of published estimates (Brodeur et al. 2016, Ioannidis, Stanley and Doucouliagos 2017, Brodeur, Cook and Heyes 2020, Ferraro and Shukla 2020). Despite growing awareness, evidence on what drives exaggeration at the study level, and therefore actionable guidance to mitigate it in non-experimental research, remain scarce (Altoè et al. 2020, Black et al. 2022). The present paper identifies concrete design parameters that can lead to exaggeration.

While the analyses from this paper are specific to studies on the acute health effects of air pollution, they can provide lessons for other types of non-experimental studies. First, a large literature investigates the short-term impacts of air pollution on different outcomes such as criminality, cognitive skills, and productivity (Herrnstadt et al. 2021, Ebenstein, Lavy and Roth 2016, Adhvaryu, Kala and Nyshadham 2022, for instance). These studies use data with a very similar structure, only focusing on different outcomes, and find effects of comparable magnitude or even smaller than those in the literature of interest. My results are therefore most likely directly applicable to these literatures. More broadly, since the impact of the drivers I identified can be explained theoretically, these drivers would affect power and exaggeration in other settings as well. A limited effect size, effective sample size, number of exogenous shocks, average count of the outcome, or strength of the instrument likely create exaggeration across a breadth of other contexts. While some design features such as a large sample size or F-statistic may seem reassuring, they do not ensure a large statistical power or protect against exaggeration and may even hide these issues. Paying careful attention to statistical power and the drivers documented in the present paper should help avoid exaggeration.

I thus advocate for systematically computing and reporting retrospective power calculations when running a non-experimental study to gauge the likelihood of exaggeration. They are easy to implement and allow us, applied researchers, to evaluate whether our research design enables us to confidently estimate a credible range of effect sizes. I implemented such calculations in Section 2 and illustrate this approach in more detail in Appendix C for an example published study. These analyses rest on hypothesized true effect sizes. Plausible values can be informed by prior studies, meta-analyses, or theoretical predictions. I do not, however, recommend reporting a Minimum Detectable Effect (MDE). An obtained estimate larger than the MDE could give a false sense of quality, since the true effect might still be smaller than the MDE. These analyses can also be complemented by prospective power calculations in the form of simulations similar to the one implemented and described in Section 3. Both of these exercises can identify the features of the design that limit power and whether adjustments, such as collecting additional data, focusing on a different subpopulation, or looking for a stronger instrument, could improve it. When such adjustments are not feasible, the calculations still help to interpret the obtained estimates, by making explicit which effect sizes the design can accurately recover and which it cannot. As an even simpler first check for potential exaggeration, large confidence intervals verging on zero can be read not only as a sign of uncertainty regarding the exact magnitude of the effect but also of potentially limited power and exaggeration of the obtained point estimate. Reporting power calculations alongside results would complement standard significance tests and provide readers with a more complete picture of the reliability of the estimates. Appendix E presents these steps as a structured workflow.

More generally, published estimates only suffer from exaggeration in the presence of publication bias. Adopting a different view of statistically insignificant results would yield important benefits (Ziliak and McCloskey 2008, Wasserstein and Lazar 2016, McShane et al. 2019). Instead of mainly interpreting point estimates, we could describe compatibility intervals: the range of effect sizes supported by the data (Amrhein, Trafimow and Greenland 2019, Romer 2020). Using such measures would better acknowledge the actual uncertainty in the estimates and better serve policy-makers.

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A Selection of the Articles from the Literature

The corpus of relevant causal articles literature was retrieved through a manual search strategy on Google Scholar, IDEAS, and PubMed. The resulting set of studies is composed of articles available before November 2022. The recent literature on the effects of air pollution on COVID-19 health outcomes is excluded to gather a relatively homogeneous corpus of studies. The list of studies selected is as follows.

Instrumental Variable Design: Moretti and Neidell (2011), Ebenstein, Frank and Reingewertz (2015), Schwartz et al. (2015), Arceo, Hanna and Oliva (2016), He, Fan and Zhou (2016), Knittel, Miller and Sanders (2016), Schlenker and Walker (2016), Sheldon and Sankaran (2017), Schwartz, Bind and Koutrakis (2017), Zhong, Cao and Wang (2017), Barwick et al. (2018), Hanlon (2018), Schwartz, Fong and Zanobetti (2018), Halliday, Lynham and de Paula (2019), Deryugina et al. (2019), Aguilar-Gomez (2020), Cheung, He and Pan (2020), Fan and Wang (2020), He, Liu and Zhou (2020), Giaccherini, Kopinska and Palma (2021), Godzinski and Suarez Castillo (2021), Guidetti, Pereda and Severnini (2021), Kim (2021), Liu and Ao (2021), Xia et al. (2022)
Reduced-Form Design: Bauernschuster, Hener and Rainer (2017), Jans, Johansson and Nilsson (2018), Jia and Ku (2019), Godzinski, Castillo et al. (2019)
Regression Discontinuity Design: Chen, Guo and Huang (2018), Fan, He and Zhou (2020), Anderson, Hyun and Lee (2022)
Event-Study Design: Mullins and Bharadwaj (2015), Simeonova et al. (2021)
Matching Design: Baccini et al. (2017), Forastiere, Carugno and Baccini (2020)

To gather a corpus of relevant articles for the epidemiology corpus, I ran the following search query on PubMed and Scopus:

```
'TITLE(("air pollution" OR "air quality" OR "particulate matter" OR "ozone"', 'OR
"nitrogen dioxide" OR "sulfur dioxide" OR "PM10" OR "PM2.5" OR', ' "carbon dioxide" OR
"carbon monoxide")', 'AND ("emergency" OR "mortality" OR "stroke" OR "cerebrovascular"
OR', '"cardiovascular" OR "death" OR "hospitalization")', 'AND NOT ("long term" OR
"long-term")) AND "short term"'
```

I retrieve the abstracts of 1,834 articles on September 28, 2021.[6] Conveniently, the common practice in these fields is to report point estimates and confidence intervals for the main results in the abstracts, allowing me to automatically retrieve information on the main results of a paper using regular expressions (RegEx). My algorithm, available online, detects phrases such as "95% confidence interval (CI)" or "95% CI" and looks for numbers directly before this phrase or after and in a confidence interval-like format. I illustrate the outcome of this procedure (in blue) using one sentence of a randomly selected article from this literature review:

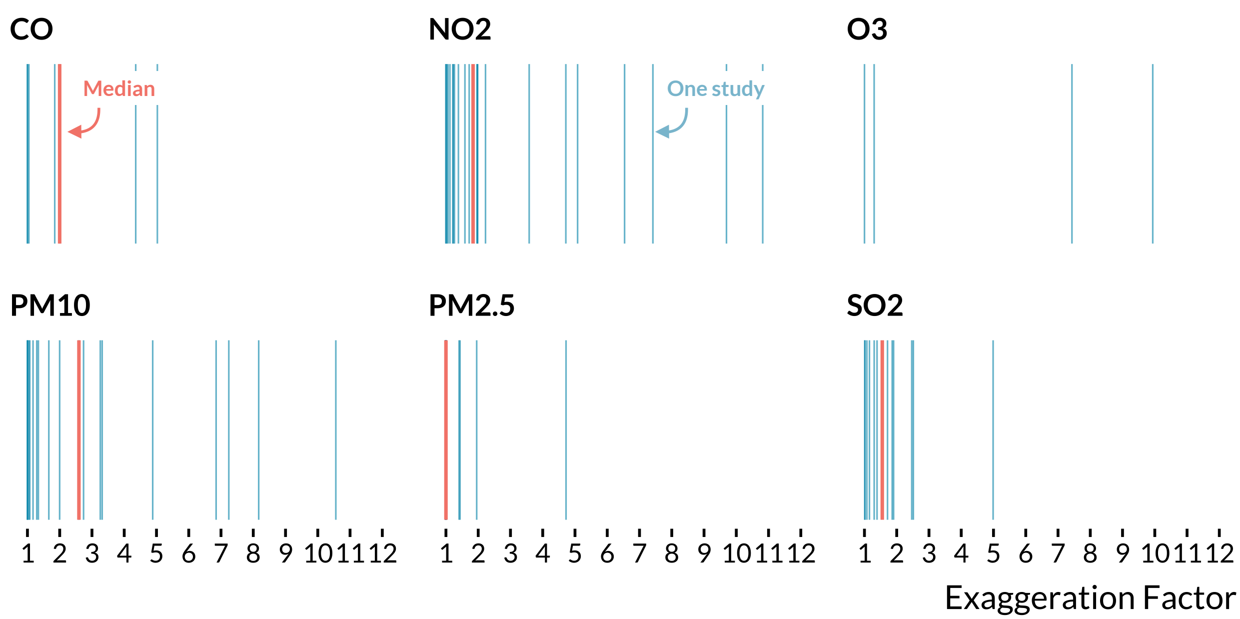
[6]The fulltext package underlying this retrieval has since been archived, preventing further updates.

"The excess risk for non-accidental mortality was **1.3% [95% confidence interval (CI), 0.8–1.7]** per 10 μg/m^3 of PM10, with higher excess risks for cardiovascular and above age 65 mortality of **1.9% (95% CI, 0.8–3.0)** and **1.5% (95% CI, 0.9–2.1)**, respectively."

Using this method, I retrieve 2,666 estimates from 784 abstracts. I then read each abstract and exclude articles whose topic falls outside the scope of our review, yielding a final corpus of 668 articles and 2,155 estimates. The sample is limited to articles that report confidence intervals and point estimates in their abstracts. I also use regular expressions to extract additional information from the abstracts, such as the pollutant and health outcome studied, the study duration, and the number of cities considered. I implement a retrospective power analysis for the 1,982 statistically significant estimates in this corpus.

B Additional Figures for the Literature Analysis

Figure B.1: Distribution of Exaggeration Ratios for Studies in Shah et al. (2015)'s Meta-Analysis

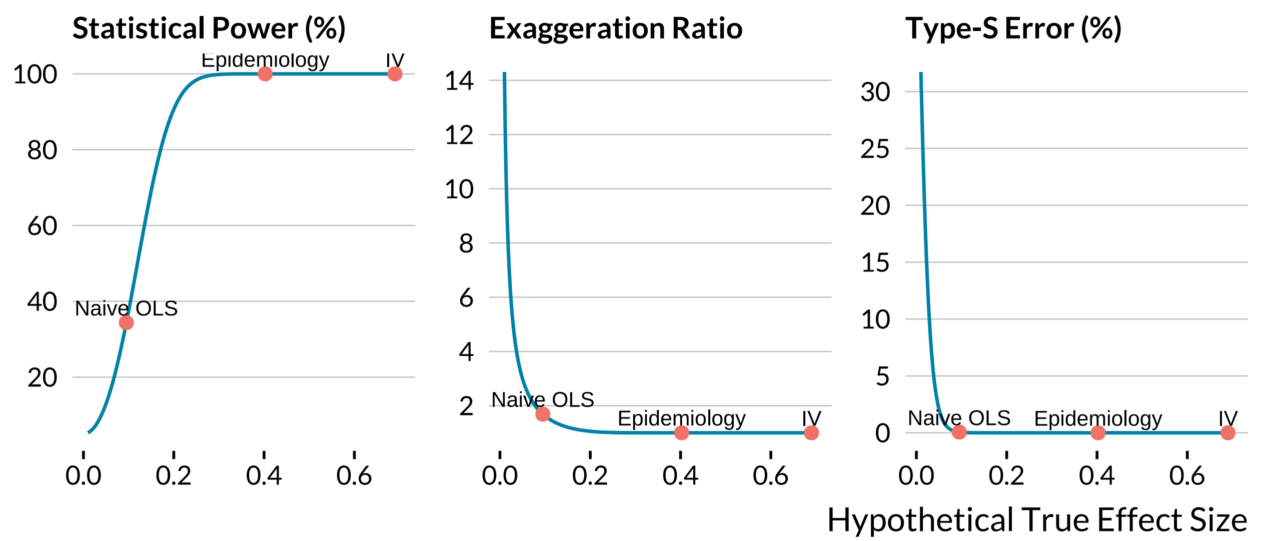


Notes: Each blue line is the exaggeration ratio of a statistically significant estimate retrieved from Shah et al. (2015)'s meta-analysis. The meta-analysis estimates are used as true effect sizes in the retrospective power calculations. Orange lines are the medians. Extreme exaggeration ratios are removed for visual clarity. The median for O_3 is 13.4.

C Implementing a Retrospective Power Analysis

I explain here how to easily implement a retrospective power analysis once a study is completed. In a flagship publication, Deryugina et al. (2019) instrument $PM_{2.5}$ concentrations with wind directions to estimate its effect on mortality, health care use, and medical costs among the US elderly. They gathered 1,980,549 daily observations at the county-level over the 1999–2013 period; it is one of the biggest sample sizes in the literature. When the authors instrument $PM_{2.5}$ with wind direction, they find that "a 1 μg/m^3 (about 10 percent of the mean) increase in $PM_{2.5}$ exposure for one day causes 0.69 additional deaths per million elderly individuals over the three-day window that spans the day of the increase and the following two days". The estimate's standard error is equal to 0.061. In Figure C.2, I plot the statistical power, the inflation factor of statistically significant estimates and the probability that they are of the wrong sign as a function of hypothetical true effect sizes.

Figure C.2: Power, Exaggeration and Type S Errors Curves for Deryugina et al. (2019)



Notes: In each panel, a metric, such as the statistical power, the exaggeration ratio or the probability of making a Type S error, is plotted against the range of hypothetical effect sizes. The "IV" label represents the value of the corresponding metric for an effect size equal to Deryugina et al. (2019)'s two-stage least squares estimate. The "Epidemiology" label stands for the estimate found in Di et al. (2017), which is the epidemiology article most similar to Deryugina et al. (2019). The "Naive OLS" label corresponds to the estimate found by Deryugina et al. (2019) when the air pollutant is not instrumented.

The estimate found by Deryugina et al. (2019) represents a relative increase of 0.18% in mortality. I label it as "IV" in Figure C.2. Is this estimated effect size large compared to those reported in the standard epidemiology literature? I found a similar article to draw a comparison. Using a case-crossover design and conditional logistic regression, Di et al. (2017) find that a 1 μg/m^3 increase in $PM_{2.5}$ is associated with a 0.105% relative increase in all-cause mortality in the Medicare population from 2000 to 2012. The effect size found by Deryugina et al. (2019) is larger than this estimate labeled as "Epidemiology" in Figure C.2. If the estimate found by Di et al. (2017) was actually the true effect size of $PM_{2.5}$ on elderly mortality, the study of Deryugina et al. (2019) would have enough statistical power to avoid exaggeration and Type S errors. Now, suppose that the true effect of the increase in $PM_{2.5}$ was 0.095 additional deaths per million elderly individuals—the estimate the authors found with a "naive" multivariate regression model. The statistical power would be 34%, the probability of making a Type S error could be null but the exaggeration factor would be on average equal to 1.7. Even with a sample size of nearly 2 million observations, Deryugina et al. (2019) could lead to a non-negligible exaggeration if the true effect size was the naive ordinary least squares estimate. Yet, the authors could argue that their instrumental variable strategy leads to a higher effect size as it overcomes unmeasured confounding bias and measurement error. Besides, for effect sizes down to 0.182 additional deaths per million elderly individuals (a 0.05% relative increase), their study has a very high statistical power and would not run into substantial exaggeration. A retrospective analysis is thus a very convenient way to think about the statistical power of a study to accurately detect alternative effect sizes.

D Case Studies

The main simulation results help understand how the various parameters influence the statistical power of studies. Yet, these parameters may not perfectly represent actual studies as I made several conservative assumptions: relatively large sample size, proportion of treated units, average outcome counts and instrumental variable strength. For each research design, I therefore consider a realistic set of parameters based on an example from the literature. I then vary the value of key parameters. As I am working with different data, I cannot exactly reproduce the level of precision found in the articles considered. My goal is not to claim that the estimates produced by a particular article are inflated, but instead to understand how low power issues could arise for representative parameter values.

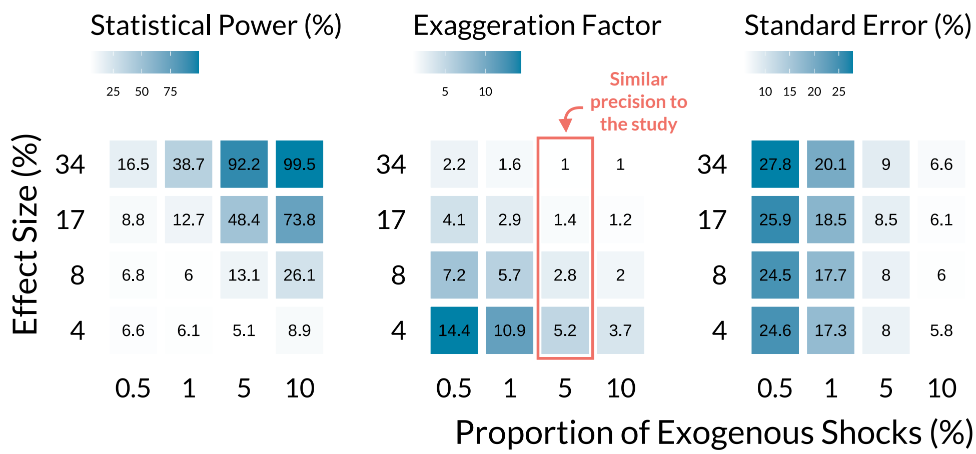
Public Transportation Strikes

Public transportation strikes are unique but rare positive shocks to air pollution as individuals use their cars to reach city centers. Even in a large data set, with several cities and a long study period, the proportion of affected days might be very small. For instance, Bauernschuster, Hener and Rainer (2017) investigate the effect of public transportation strikes on air pollution and emergency admission in the five biggest German cities over a period of

6 years. Despite a sample size of 11,000, there are only 57 1-day strikes during the study period (0.5% of days are actually treated). The authors find that children hospitalizations for breathing issues increase by 34% (SE=8%) on strike days. On average, 0.22 children per day go to the hospital for breathing issues.

I simulate a similar design with my own data. I first randomly sample 2200 observations for five cities and then vary (i) the proportion of exogenous shocks from 0.5% up to 10%, and (ii) the treatment effect size from a 4% increase up to a 34% increase. I focus on elderly mortality due to chronic obstructive pulmonary disease since it has an average daily count of 0.29 cases.

Figure D.3: Evolution of Power and Exaggeration for Public Transportation Strikes Designs



Notes: Each panel displays the average value of a metric (power, exaggeration, and standard error) for varying proportions of exogenous shocks and effect sizes. The average standard error of simulations is the raw standard error divided by the mean number of cases of the health outcome. For each combination of parameters, 1000 simulations were run.

Figure D.3 displays the simulation results. The first panel from the left shows that both large effect sizes and a large proportion of exogenous shocks are required to reach adequate power. The middle panel shows that a proportion of 0.5% of exogenous shocks is associated with very large exaggeration ratios, from 2.2 for a true effect size of 34% up to 14 for one of 4%. Power issues fade for a combination of a proportion of exogenous shocks above 5% and effect sizes above 17%. The right panel plots the average standard error of the estimates, expressed as a fraction of the average of the health outcome. Bauernschuster, Hener and Rainer (2017) reports a standard error of 8%. In the simulations, we recover that specific precision for a proportion of exogenous shocks of 5%. In that case, a true effect size of 34% would not yield inflated estimates. However, if effect sizes are actually smaller and more representative of those found in the literature, the exaggeration would be consequential.

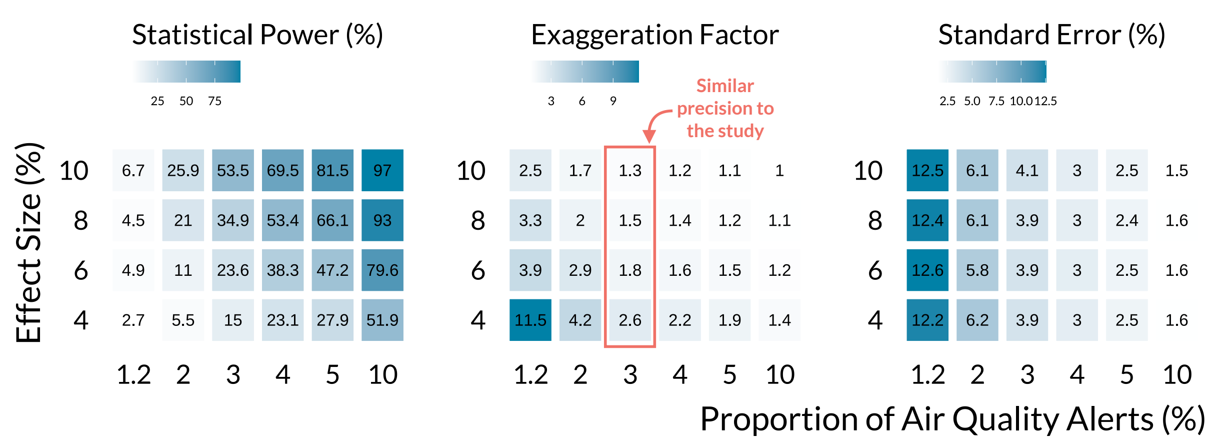
This simulation exercise shows that exaggeration is likely to arise in practice since the proportion of exogenous shocks is low. It occurs even when true effect sizes are relatively large.

Air Pollution Alerts

Air pollution alerts are also rare events. Their effects are estimated using regression discontinuity designs that restrict the analysis to observations close to the air quality threshold. As a consequence, the effective sample size may be particularly small. For instance, Chen et al. (2018) investigate the effects of air quality alerts on emergency department visits in Toronto, over the 2003-2012 period. While the nominal sample size is 3652, the effective one is only 143 (100 control days and 43 treated days). Only 1.2% of observations are treated. The authors find that eligibility to air quality (the intention-to-treat effect) approximately reduces emergency visits for asthma by 8% (SE=3.8%). The average daily count of cases of their health outcome is 26.

I approximate the setting of Chen et al. (2018) using my data. I first sample one city for a time period of 3652 days and randomly allocate the treatment. I then repeat the process varying the proportion of alerts and effect sizes. My outcome variable is the total number of non-accidental deaths since it has a daily mean of 23.

Figure D.4: Evolution of Power and Exaggeration for Air Quality Alerts Designs



Notes: Each panel displays the average value of a metric (power, exaggeration, and standard error) for varying proportions of exogenous shocks and effect sizes. The average standard error of simulations is the raw standard error divided by the mean number of cases of the health outcome. For each combination of parameters, 1000 simulations were run.

Figure D.4 displays the simulations results. As in Figure D.3, a combination of large effect sizes and many air quality alerts is needed to avoid low power issues. We get a precision similar to Chen et al. (2018) for a proportion of air quality alerts of 3%. For an effect size of 4%, the average exaggeration ratio is equal to 2.6. In that case, the average of statistically significant estimates is 10%, a magnitude similar to that found by Chen et al. (2018), thus suggesting that the effect found by the study could be exaggerated.

Unless true effect sizes are very large, air quality alert designs produce inflated estimates in realistic settings.

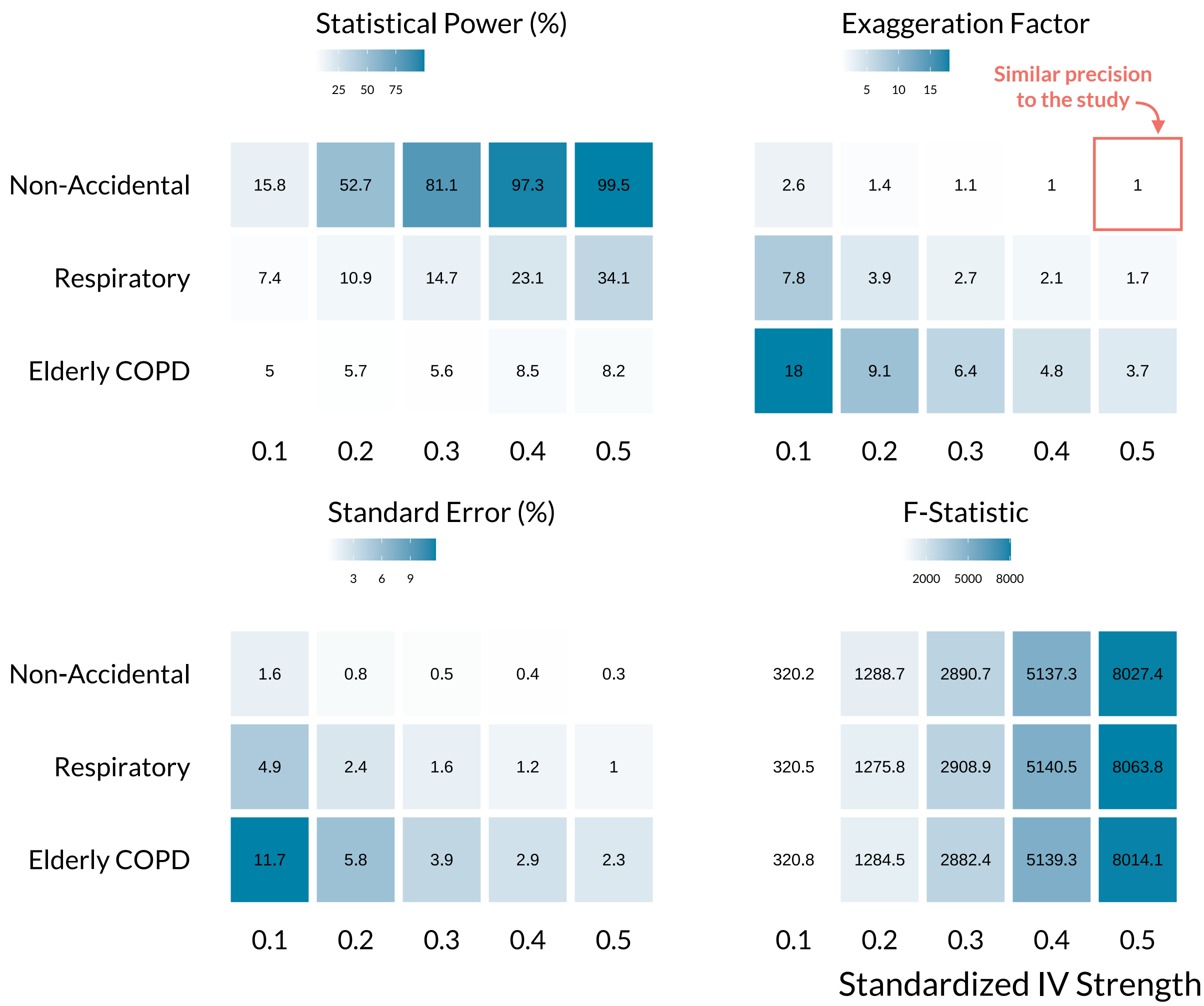
Instrumenting Air Pollution

Finally, I investigate the most commonly used strategy in the causal inference literature, the instrumental variable design. Several studies rely on very large datasets and exploit changes in weather patterns as sources of exogenous variations. For instance, Schwartz, Fong and Zanobetti (2018) instrument $PM_{2.5}$ concentration with planetary boundary layer, wind speed, and air pressure. Once the effects of seasonal and other weather parameters are accounted for, the combination of their instruments explains 18% of the variation in $PM_{2.5}$ concentration. They find that a 10 $\mu g/m^3$ increase in $PM_{2.5}$ leads to a 1.5% (SE=0.22%) increase in daily non-accidental mortality. There are on average 23 daily deaths in their dataset of 591,570 observations (135 cities with a length of study of approximately 4382 days).

In my simulations, I assess how the strength of the instrumental variable affects power issues for several health outcomes. I consider a binary instrumental variable and vary its effect on air pollution concentration from a 0.1 to a 0.5 standard deviation increase. The 18% correlation in Schwartz, Fong and Zanobetti (2018) corresponds to a 0.4 standard deviation increase in this case (Lipsey and Wilson 2001). I assume that half of the observations are exposed to exogenous shocks. I set an effect size corresponding to a 1.5% relative increase in three health outcomes with different average number of cases: non-accidental mortality (mean cases of 23), respiratory mortality (mean of 2), and chronic obstructive pulmonary mortality of elderly (mean of 0.3). As my data set is smaller than the one used in Schwartz, Fong and Zanobetti (2018), I only run simulations for a sample size of 100,000.

In Figure D.5, the top-left panel shows that power reaches a satisfactory level for large instrumental variable strengths but only for non-accidental causes. For respiratory and elderly mortality, exaggeration can be substantial even for large IV strength. While the sample size is large, it is smaller than the one in Schwartz, Fong and Zanobetti (2018). As a consequence, the simulations only have a precision close to theirs for an instrumental variable strength of 0.5 and non-accidental mortality. Yet, the simulations highlight that important exaggeration issues can arise in realistic settings, even for large IV strength. The bottom-right panel of Figure D.5 confirms the result found in the simulations of the previ-

Figure D.5: Evolution of Power and Exaggeration for Instrumental Variable Designs



Notes: Each panel displays the average value of a metric (power, exaggeration, standard error, and first-stage F-statistic.) for varying proportions of exogenous shocks and effect sizes. The average standard error of simulations is the raw standard error divided by the mean number of cases of the health outcome. For each combination of parameters, 1000 simulations were run.

ous section: a large first stage F-statistic can be a poor indicator of statistical power issues. For instance, for non-accidental mortality and an IV strength of 0.1, the F-statistic is equal to 320 but the exaggeration factor is 2.6, with an associated estimated power of 16%. Importantly, as the F-statistic does not vary with the number of cases in the outcome it can all the more hide important power issues.

E A Workflow to Evaluate Exaggeration Risk

The analyses in this paper suggest the following workflow for evaluating exaggeration risk in a non-experimental study. The first three steps apply at the design stage; the fourth is also useful retrospectively, once a study is completed.

1. **Identifying a range of plausible effect sizes**. Draw on prior studies, meta-analyses, theoretical predictions, or, when comparing strategies, on alternative estimators of the same quantity such as the OLS analogue of an IV estimate. Section 2 provides a more thorough discussion of the various options.

2. **Running a simulation**. Before conducting a study, implement a simulation by generating data from scratch or using an existing dataset different from the one used in the study. One may want to proceed incrementally, starting with a simple and probably unrealistic setting before making it more complex. Then, estimate the model of interest on the generated dataset, and compute the power, exaggeration ratio, and rate of Type S error. Section 3 and the project's website provide more details on how to run a simulation.

3. **Identifying the design features that limit power**. Vary the parameters of the simulation and compute the metrics of interest over an array of values. The drivers documented in Section 3 rarely appear in isolation, and large sample sizes or first-stage F-statistics do not protect against any of them. Decide where to best invest resources to reach sufficient power to accurately recover the plausible effect sizes from step 1.

4. **Running retrospective power calculations**, using closed-form solutions, implemented in the `retrodesign` package for instance. Discuss which of the plausible effect sizes the design can accurately recover and which it cannot, and interpret results accordingly. Section 2 provides more details on how to run such an analysis and Appendix C describes how to implement it for an example study.